

Mobility, maps & travel

BENGALURU, FRIDAY 6:40 PM, HEAVY RAIN ON OUTER RING ROAD

Ananya, 29, has finished work in Bellandur and wants to get home to Jayanagar, 14 km away. Her cab app says “finding your driver” for the fourth time. Two drivers accepted her ride and cancelled within a minute. The fare now shows 1.8× surge, ₹610 instead of ₹340. A bike taxi is ₹190, but it is raining. Across the city, thousands of riders are staring at the same spinning wheel, while hundreds of drivers sit in traffic with a passenger they picked up 40 minutes ago.

Try it: drivers keep cancelling tonight. Is this a pricing problem, a matching problem or a driver problem, and how would you tell which? Think for five minutes before reading on.

How a ₹300 cab ride pays three people

A ride-hailing app sells one thing: a car or bike that arrives in a few minutes, at a price you expected, and takes you there safely. It owns no cars. It rents the time of drivers who own or lease their vehicles, and it earns either a slice of each fare or a flat daily fee. Everything a PM does in this sector comes down to one balance: enough drivers near enough riders, at a price both accept. Maps and travel apps sit on the same foundations: knowing where everyone is, how long things take, and who will pay for what.

3.9 bn

Uber trips worldwide in April–June 2026, up 18%; India is one of its largest markets by trips

₹934 Cr

Rapido's revenue in FY25, up 44%; valued at \$3 bn in May 2026

2×

the most an app may multiply the base fare at peak, under India's 2025 aggregator guidelines

\$10.4 bn

travel booked on MakeMyTrip in FY26 (to March 2026); its revenue passed \$1 bn

Who does what, and who pays whom

Five parties touch every ride. The **rider** pays the fare, by UPI, card or cash. The **platform** (Uber, Ola, Rapido, Namma Yatri) runs the app, sets the price, matches the ride and handles safety and support. The **driver** drives, and usually owns the vehicle or pays an EMI on it. Some drivers work for a **fleet owner** who owns 20 or 200 cars and pays them a daily wage or a share. And behind every screen sits a **maps provider** (Google Maps, or the platform's own map, like Ola Maps) that turns addresses into routes, times and prices. Figure 4.1 follows the money on one ride.

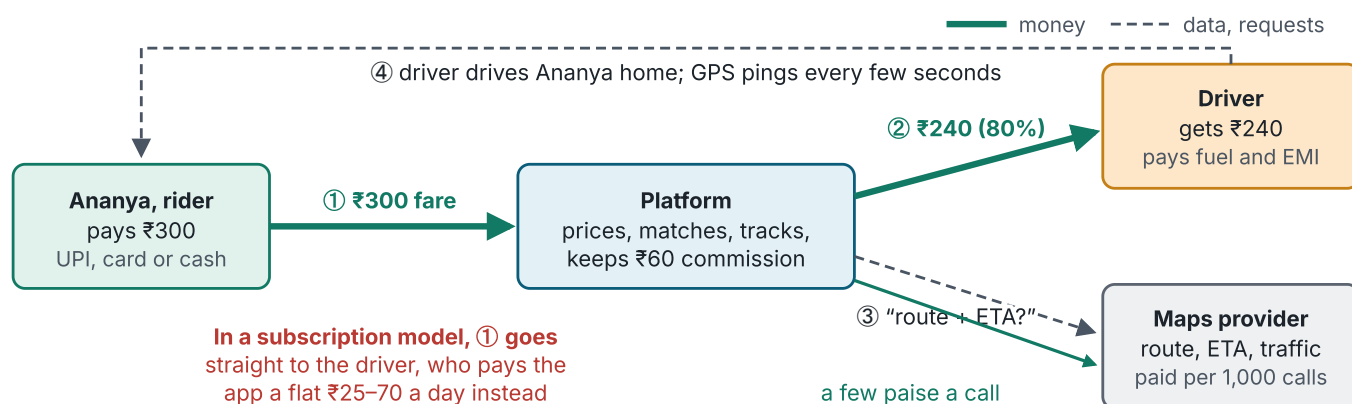


Figure 4.1 · Who pays whom on one ride. Under the commission model the platform keeps a slice of every fare; India's 2025 guidelines cap that slice at 20% when drivers use their own vehicles. The maps bill is small per ride but large across a billion rides. Illustrative ₹300 cab ride; 80% driver share from the Motor Vehicle Aggregator Guidelines, July 2025.

The unit economics, worked in rupees

Take Ananya's ₹300 ride from the **driver's side**. The trip is 12 km, and the driver drove 3 km empty to reach her. A CNG hatchback costs about ₹5 a km to run, fuel and tyres and servicing together: 15 km × ₹5 = ₹75. The commission is 20%: ₹60. The car's EMI is about ₹15,000 a month; spread over 26 working days and 12 trips a day, that is about ₹48 a trip. The driver keeps about ₹117 (Figure 4.2).

Driver's side of one ₹300 cab ride
(12 km trip + 3 km to the pickup)

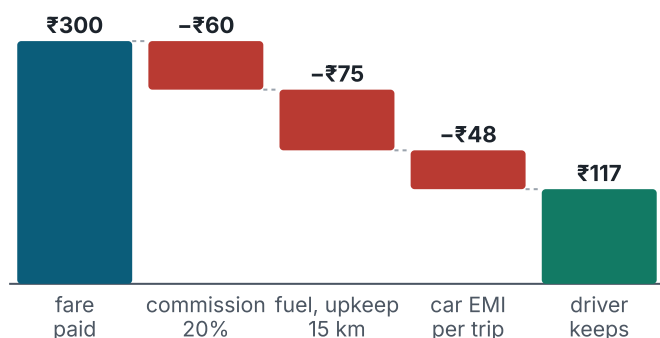


Figure 4.2 · A ₹300 ride, for the driver. Only about ₹117 of the fare is the driver's income; fuel and the car loan take more than the app does. *Illustrative; CNG running cost ~₹5/km, 20% commission, ₹15,000 EMI.*

Platform's side of the same ride:
₹60 of commission shrinks fast

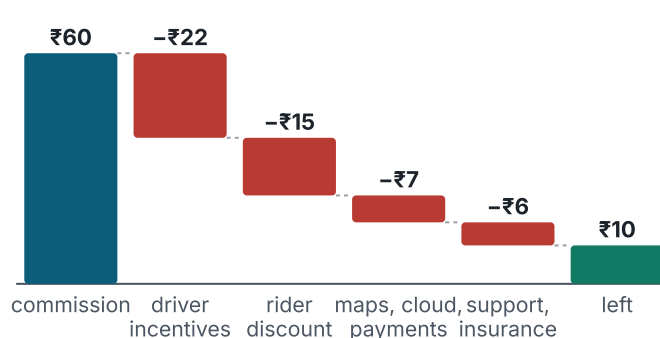


Figure 4.3 · The same ride, for the platform. Driver bonuses and rider discounts eat most of the ₹60. In a price war this bar goes below zero. *Illustrative; categories typical of ride-hailing accounts.*

Now add time. A driver who is online 11 hours and completes 12 trips of about 35 minutes each spends $12 \times 35 = 420$ minutes, 7 hours, with a rider in the car. That share, $7 \div 11 = 64\%$, is **utilisation** (the share of online time spent carrying a paying rider). The driver earns $12 \times ₹117 = ₹1,404$ a day, or about ₹128 per online hour. If utilisation falls to 50% (fewer trips, longer waits), earnings per hour fall by a fifth, and the driver logs off or switches app. **Utilisation is the number that decides whether drivers stay, and drivers decide whether riders get a car.**

Commission or subscription: the fight of 2025–26

For a decade apps took 20–30% of each fare. Then Namma Yatri (Bengaluru, 2022) and Rapido charged drivers a small flat fee instead: Namma Yatri asks ₹25 a day for unlimited rides. Drivers loved it. Uber moved autos to a subscription in February 2025, and Ola announced “zero commission” in June 2025. Figure 4.4 shows why the switch is so attractive to drivers and so painful for platforms.

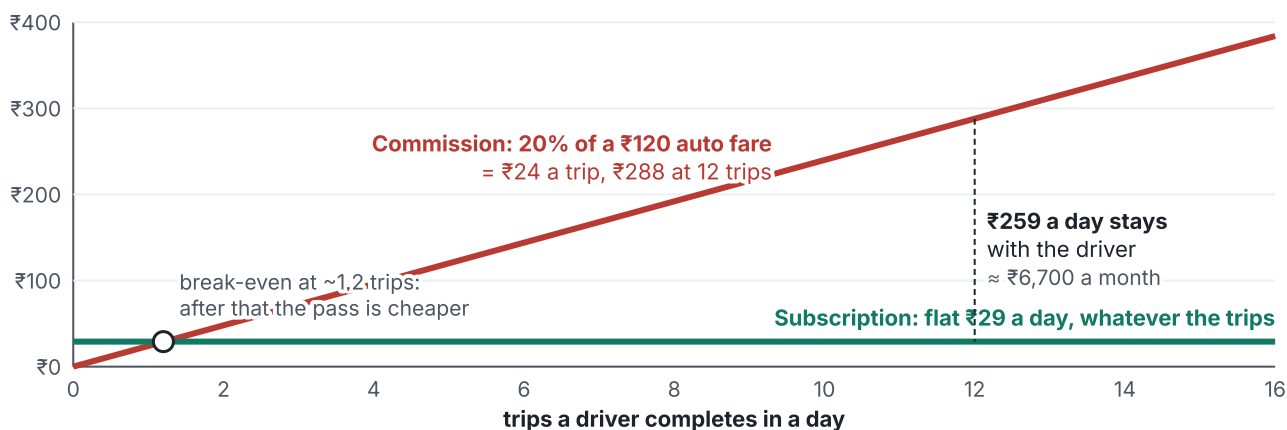


Figure 4.4 · What the platform takes from one auto driver in a day. Past about one trip a day, the flat pass is cheaper for the driver. At 12 trips the driver keeps ₹259 more a day, and the platform earns a tenth of what it did. That is why subscription apps must earn elsewhere: ads, loans for vehicles, insurance, delivery. *Illustrative ₹120 average auto fare; ₹29 daily pass, within the ₹25–70 range charged by Namma Yatri, Rapido and Ola in 2025–26.*

The subscription model is **counter-positioning** (a move a rival can copy only by hurting its own business): Uber could match Rapido's cheaper autos only by giving up commission income it depended on. It did, for autos, and kept commission on cabs, where riders pay more and drivers have fewer alternatives.

The metrics that matter

Metric	What it means, in one line	Why a PM watches it
Completed trips	Rides that ended at the destination, paid	The usual North Star: both sides got what they came for
ETA to pickup	Minutes from booking to the car arriving	Riders leave when it passes 6–8 minutes; it measures supply nearby
Request-to-completion rate	Completed trips ÷ ride requests	Catches every failure: no driver found, cancellations, no-shows
Acceptance and cancellation rates	Share of offers drivers accept; share of accepted rides cancelled, by driver or rider	The first sign that price, distance or payment mode is wrong for drivers
Utilisation	Time with a rider ÷ time online	Drives driver earnings, and so driver supply
Earnings per online hour	What a driver takes home for each hour logged in	The number drivers compare across apps
Surge share	Share of trips priced above the base fare	Too high and riders feel cheated; too low and supply dries up at peaks
Trips per rider per month	How often a rider comes back	Habit: commuters ride 20+ times a month, occasional riders twice

Players and numbers

Share of rides by vehicle type, India, 2025 (approximate)

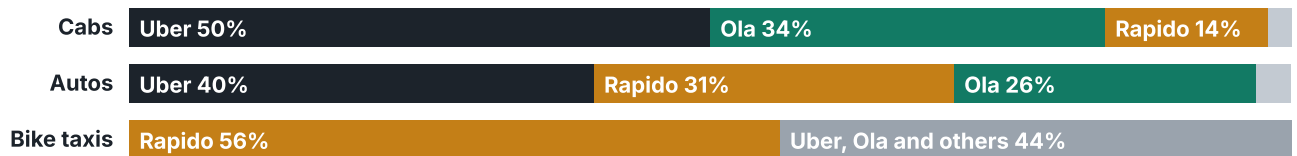


Figure 4.5 · Who leads each kind of ride. Uber leads cabs and autos; Rapido leads bikes and is the fastest-growing challenger in autos. Ola, once the market leader, has lost ground everywhere. *Approximate shares compiled from industry estimates (Equentis and press reports, 2025); they vary widely by city.*

Player	What it is	Latest numbers (dated)
Uber	Global leader; cabs on commission, autos on subscription in India; Uber One membership	Worldwide Q2 2026: gross bookings \$58.0 bn (+22%), 208 mn monthly users, adjusted EBITDA \$2.8 bn
Ola	India's first big ride app (2010); built Ola Maps in 2024; moved to zero commission in 2025	FY25 revenue ₹1,171 Cr (–42%), loss ₹662 Cr; preparing an IPO (2026 filings)
Rapido	Started with bike taxis (2015); driver subscriptions; now autos, cabs, parcels and food (Ownly)	FY25 revenue ₹934 Cr (+44%), loss ₹258 Cr; raised \$240 mn at \$3 bn (May 2026)
Namma Yatri	Driver-first app built on open ONDC/Beckn rails; ₹25 a day, no commission	7.7 lakh drivers enabled (Aug 2026)
BluSmart	All-electric cabs, salaried drivers, owned fleet	~8,000 EVs; stopped in April 2025 after SEBI's order on diverted EV loans
Google Maps · Mappls	Maps, routes, traffic, places; Mappls (MapmyIndia) is India's homegrown map	Google Maps: 2 bn+ monthly users; Gemini-powered "Ask Maps" launched in India, March 2026
MakeMyTrip	India's largest online travel agency: flights, hotels, holidays, buses (redBus), trains	FY26 gross bookings \$10.4 bn, revenue \$1.04 bn, operating profit \$156 mn
ixigo · EaseMyTrip · IRCTC	Trains and budget flights; low-cost air bookings; the railways' own ticketing site	All listed in India; IRCTC sells most train tickets booked online

Three turning points

The fight moved from “who has more riders” (discounts) to “who do drivers prefer” (fees), then to “what do the rules allow”.

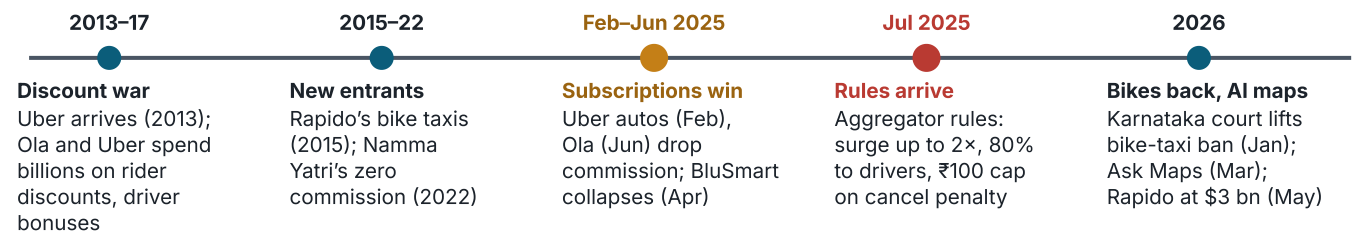


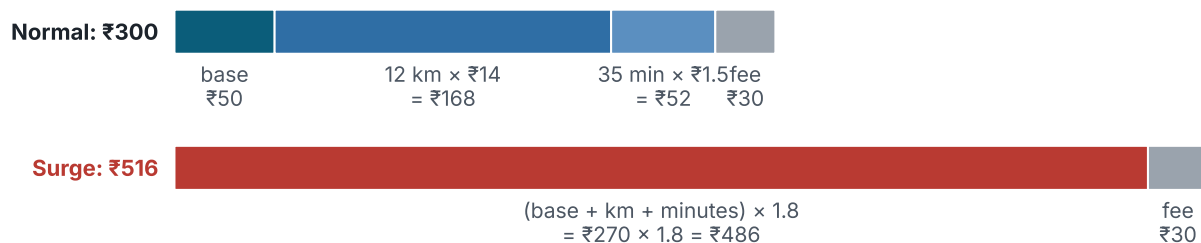
Figure 4.6 · How Indian ride-hailing changed. Each turning point shifted power towards drivers or regulators, and away from the platform's ability to set prices freely. *Company announcements; MoRTH, July 2025; Karnataka High Court, 23 January 2026 (the state has appealed to the Supreme Court).*

1. The discount war (2013–17). Uber and Ola each paid riders to ride and drivers to drive. Riders learned that a cab could cost less than an auto. When the money stopped, prices rose and drivers' bonuses vanished, and the driver strikes of 2018 followed. The lesson: subsidies buy trips, not loyalty. **2. Drivers get choice (2022–25).** Namma Yatri and Rapido showed that a platform could charge a flat fee, and Uber and Ola had to follow for autos. **3. The rules arrive (2025).** The Motor Vehicle Aggregator Guidelines let states cap surge at 2× the base fare, set a floor at 50% off-peak, give drivers at least 80% of the fare on their own vehicles, and allow a 10% cancellation penalty (up to ₹100) on whoever cancels without a valid reason. BluSmart's collapse the same year, after SEBI found EV loans had been diverted, ended the idea that an owned, salaried fleet was the safe alternative.

How the fare is built

Most apps build a fare from four parts: a **base fare** for starting the trip, a rate per **kilometre**, a rate per **minute** (so a driver stuck in traffic is still paid), and a **platform or booking fee**. Ananya's normal-hour fare: ₹50 + 12 km × ₹14 + 35 min × ₹1.5 + ₹30 = ₹50 + ₹168 + ₹52 + ₹30 = ₹300. At 1.8× surge the first three parts are multiplied: (₹50 + ₹168 + ₹52) × 1.8 = ₹486, plus the ₹30 fee = ₹516 (Figure 4.7). Apps now show this **upfront fare** (a fixed price quoted before booking) instead of running a meter, so the app, not the rider, carries the risk of a traffic jam.

How a ₹300 fare is built (normal hour), and the same ride at 1.8× surge



The 2025 guidelines let states cap the multiplier at 2× the base fare; here the fee sits outside the multiplier.

Figure 4.7 · The four parts of a fare, and what surge multiplies. A surge multiplier scales the whole distance-and-time price, which is why a long ride at 1.8× hurts far more than a short one. *Illustrative rate card; actual rates differ by city, vehicle type and app.*

Five mechanics that aren't obvious

1. Drivers cancel because they can see the maths. A driver who accepts a ride and then learns the drop is far out, with no return trip, faces an hour of empty driving. Cash versus UPI, a short low fare and a far pickup are the other usual reasons (Figure 4.8). Showing drivers more information can *raise* cancellations.

Why a driver cancels after accepting: illustrative split of 100 driver cancellations in a metro

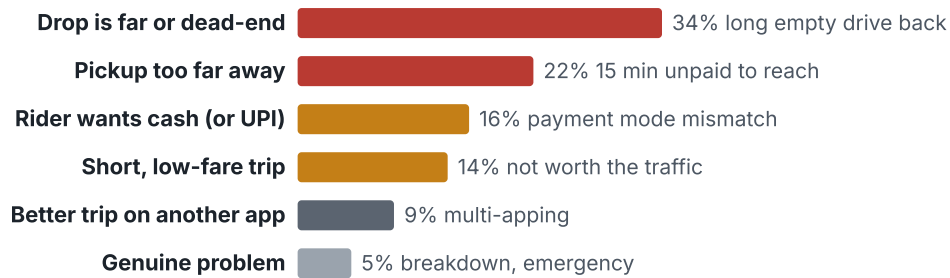


Figure 4.8 · Why drivers cancel after accepting. Most reasons are about the driver's next hour of earnings, not about the rider. Fix the economics of bad trips and many cancellations disappear. *Illustrative split, built from driver surveys and press reporting on cancellation reasons, 2024–25; real shares vary by city and hour.*

2. Surge is a signal to drivers more than a price for riders. A 1.5× price tells drivers across town where the money is tonight, and pulls them in. It also sends some riders to the metro, which frees cars for those who need them most. Capped too low, surge stops working as a signal. **3. Multi-apping.** Most drivers run two or three apps at once and take the best offer. Loyalty is earned per trip, which is why cancellation rates rise the moment a rival app pays more. **4. Airports and stations are separate markets.** Drivers queue for an hour in airport lots for one long, high-fare ride, so a city can be short of cars while 300 wait at the airport. Apps run special queues and pricing there. **5. The map is a hidden cost and a moat.** Every search, price estimate and ETA calls a maps service. Ola's founder said moving to its own Ola Maps in 2024 cut a bill of about ₹100 Cr a year to near zero. Better ETAs also mean better prices and fewer cancellations.

Travel booking in one paragraph

An **online travel agency** (OTA, such as MakeMyTrip, ixigo or Booking.com) sells other companies' seats and rooms and keeps a cut. The cut depends on how much choice the seller has. Airlines sell their own tickets and pay OTAs little, so a ₹10,000 flight earns an OTA a few hundred rupees, mostly convenience fees. A hotel, which needs help filling rooms, pays 15–20%. That is why every OTA pushes you from a flight to a hotel and a holiday package (Figure 4.9).

What an online travel agency keeps on ₹10,000 of each kind of booking (illustrative ranges)

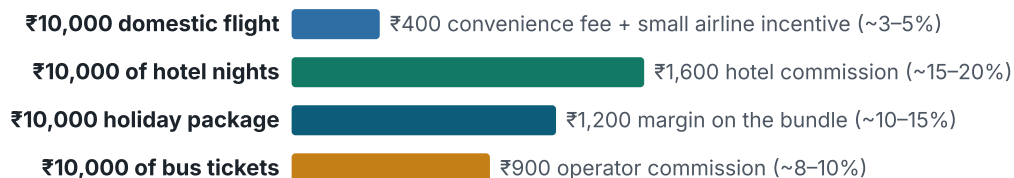


Figure 4.9 · Why travel apps want you to book the hotel. The flight brings the customer; the hotel and the package earn the money. *Illustrative ranges typical of Indian OTAs, 2025–26; MakeMyTrip reports hotels and packages as its fastest-growing profit line (FY26).*

IN THE INTERVIEW · QUESTIONS THIS SECTOR LOVES

- **“Cancellations are up. Diagnose.”** Split driver from rider cancellations first; then look at drop-off distance, pickup distance, payment mode and time of day.
- **“Should Uber move cabs to subscription?”** Compare lost commission against extra drivers and cheaper fares; pilot in one city; guard ETA and driver earnings.
- **“Is surge fair?”** Explain it as a signal that brings drivers out, then say where it fails: floods, emergencies, and when the cap makes it too weak to move supply.
- **“Launch in a new city.”** Drivers first, a few busy corridors, density before breadth.
- **“How would you improve Maps?”** Pick a user with a hard job (a tourist, a delivery rider), not “everyone”.

CASE 4.1 · OLA FOR SCHOOLCHILDREN**Product design**Medium IIM A *Product Mindset 2026–27*, pp. 250–251

Try it first. Think of how you got to school at age ten. Who chose that van or bus, who paid, and what would have made your parents switch? Now spend five minutes on what Ola would need to be true before building anything.

THE QUESTION

Ola has identified the travel needs of school children as a business opportunity. Design the experience on Ola to meet their needs.

A Would this be integrated into the existing Ola app or a new one? And what's the region of operation?

Q Integrated into the existing Ola environment in India.

A The service is for children aged 4-17, but the user and the buyer are different people. The child is the rider. The parent is the decision maker and the payer, and is the person the service actually has to convince. The school is a third stakeholder - it controls gate timings and pick-up rules, and at scale it is the cheapest channel to reach parents. The driver is the supply side. So I'll design for the parent's decision and the child's experience, and treat the school as a partner rather than a customer. I'll segment on the distance between home and school - closer students may walk or cycle, while farther students need comfortable transport and on the child's current commute mode, since that is what the service has to displace.

Q Sounds good. Which segment should the service target?

A Looking at the four modes in use today:

Current mode	Why	Target?
School-run bus	School-provided, route-fixed and cheap. Trusted by default and available to families across income levels.	No - hard to displace
Private car or family driver	The commute is already solved and the parent controls it.	No - low intent to switch
Informal shared auto or van	Unverified driver, no accountability, no receipts, frequent overcrowding. Largest group.	Yes - launch here
Walk or cycle	Distance is too short for the service to add value.	No

A This is where the pain is sharpest and where a verified, trackable alternative is worth paying for. Once density and utilisation improve, a lower-priced tier can extend the service to more price-sensitive families.

Q Fair. What other factors will you consider?

A The mode of transport, and because this is a child-safety service, I'll choose modes by safety and age-appropriateness first, not just cost:

Mode	Suitability for child transport
Bus	Secure and disciplined, but high establishment cost to add - excluded at launch
Auto	Informal and less safe; acceptable only for older students (e.g. 13+) with safeguards
Van	Enclosed, relatively safe, flexible timings - well-suited; primary mode
Cab	Enclosed and comfortable but pricier; suited to longer distances
Bike	Unsafe for children and infeasible in monsoon - excluded

A As the current Ola service already includes travel via autos, cabs, and bikes, we can target all the service segments excluding buses, as including buses could result in a huge establishment cost.

Q That's alright. What are the user pain points the service should solve?

A Three buckets:

1. Driver / vehicle selection:

- poor vehicle condition
- few driver/vehicle options
- price unrelated to service quality

2. Commute:

- driver indiscipline and irregular leaves
- information asymmetry on driving skill

- missing licences/paperwork; overcrowding

3. Payments:

- irregular price hikes
- no receipts or official paperwork

These are all must-solves to earn parents' trust and keep children safe, and the incremental effort to add each is low, so the MVP should solve all of them.

Q Okay, now let's move on to the solutions.

A The solution is an end-to-end school pick-up and drop service:

- Ola-verified drivers: driving tests and background checks, with always-on GPS tracking to monitor driving quality.
- Free trial rides: parents experience the service, meet the driver and check the vehicle before subscribing; feedback and a callback if they're unhappy.
- Real-time location updates: notifications for ride start, driver 10 minutes away, arrival at home and at school.
- Backup arrangements: a no-extra-charge backup if the regular vehicle breaks down or the driver is unavailable.
- Capacity caps (overcrowding): a fixed seat allocation per child, so vehicles are never overcrowded.
- Transparent payments: fixed subscription pricing (no surge or ad-hoc hikes) and in-app receipts, directly addressing the payments pain points. Every feature here answers an anxiety the parent has, because the parent is the one who signs up and renews. The child's side of the design is narrower - a fixed seat, a familiar driver and a ride that arrives on time.

Q Alright. Can you walk-me through the proposed MVP?

A Four steps:

1. Sign in:
 - create a profile (name, age, photograph)
 - select school and home address
2. Select service and driver:
 - view price estimates per service
 - see driver and vehicle details
 - schedule a test ride, finalise the driver
3. Payment and agreement:
 - choose a subscription (monthly/quarterly/yearly)
 - and sign the online agreement
4. Avail the service:
 - confirm the weekly schedule (driver notified of minor tweaks),
 - track live location,
 - and select a backup if needed

Q That sums up the case nicely. Thank you for your time.

WHAT WORKED · WHAT WAS MISSING

- | | |
|--|---|
| <ul style="list-style-type: none"> ✓ Separated the buyer from the user in the first answer: the parent decides and pays, the child rides, the school controls the gate, the driver is supply. ✓ Segmented by today's commute and picked the informal van, where the pain is sharpest, with a clear table of why not the others. ✓ Ruled out modes on safety first (no bikes, no autos for young children) before cost. ✓ Tied every feature to a parent's worry: verification, trial ride, alerts, backup, seat caps, fixed pricing. | <ul style="list-style-type: none"> ✗ "The MVP should solve all of them." That is the whole product. Nothing tests whether parents will pay more than they pay the van driver they know. ✗ No numbers. No price, no children per van, no cost of a dedicated driver. The idea lives or dies on route density. ✗ Ola's supply doesn't fit. Ola drivers are on-demand and use several apps. A school run needs the same driver at 7:15 am every school day. ✗ Never asked why Ola wants this: new revenue, idle afternoon supply, or brand trust with families. ✗ No success metric or safety check, even though the interviewer closed the case without asking for one. |
|--|---|

Scoping kit check: Confirm the thing ✓ (**inside the Ola app, India**) · Goal ✗ · Who ✓ (**parent, child, school, driver**) · Today and the trigger ✓ (**current commute modes**) · Edges ✗ **city, regulation**. Design add-on “eligibility and legal limits” ✗: for minors this is the first question. Most states set rules for vehicles that carry schoolchildren (markings, speed limits, sometimes an attendant), and an MVP must meet them from day one.

CHECK · “THE INCREMENTAL EFFORT TO ADD EACH IS LOW, SO THE MVP SHOULD SOLVE ALL OF THEM” DOESN'T HOLD UP

Background checks, a backup vehicle on standby, fixed seats and a dedicated driver each need a new operation, not a new screen. Ola's drivers are independent and paid per trip; a backup van waiting every morning costs a full day's wage. The right test is the opposite: find the one thing that must be true, and test only that.

STRONGER ANSWER



Figure 4.10 · A design answer that learns before it builds. Move ③ decides what the MVP must test; everything else waits. *Author's summary.*

A ① CLARIFY

Why does Ola want this: a new revenue line, work for drivers in quiet hours, or trust with families? And which city first? For children, I'd also confirm the state's rules for school vehicles, because they set the minimum design. I'll assume children aged 6–14.

Q A new, predictable revenue line. Start with Bengaluru.

Why this move: “predictable revenue” points to a monthly subscription, not per-ride pricing. The city sets traffic, school timings and rules.

A ② PICK THE FAMILIES

I agree with the informal van users: the pain is sharpest there. Illustrative size for one large school: 2,000 children × 30% in private vans = 600 children. Those parents already pay about ₹2,000–2,500 a month and worry about the unknown driver and a van packed with 12 children.

Why this move: families already paying for a van have proven they will pay for school transport. We only need them to switch.

A ③ RISKIEST ASSUMPTION

Five things must be true: parents pay more (say ₹3,000); each van fills enough seats; Ola can keep the same driver daily; children are fine with it; tracking works. The last one is easy, Ola already does it. The first two are both unknown and fatal (Figure 4.12).

The maths shows why. A van with a driver and an attendant costs about ₹40,000 a month. It can do two school runs each morning. At ₹3,000 a child, each extra child in a run adds $2 \times ₹3,000 = ₹6,000$ a month. Break-even = $₹40,000 \div ₹6,000 \approx 6.7$ children a run (Figure 4.11).

Why this move: the numbers turn a list of worries into one question: will parents pay ₹3,000 and can we fill 7 seats a run? That question is what the MVP must answer.

Monthly profit per van: 2 school runs a morning, ₹3,000 per child, costs ₹40,000 (illustrative)

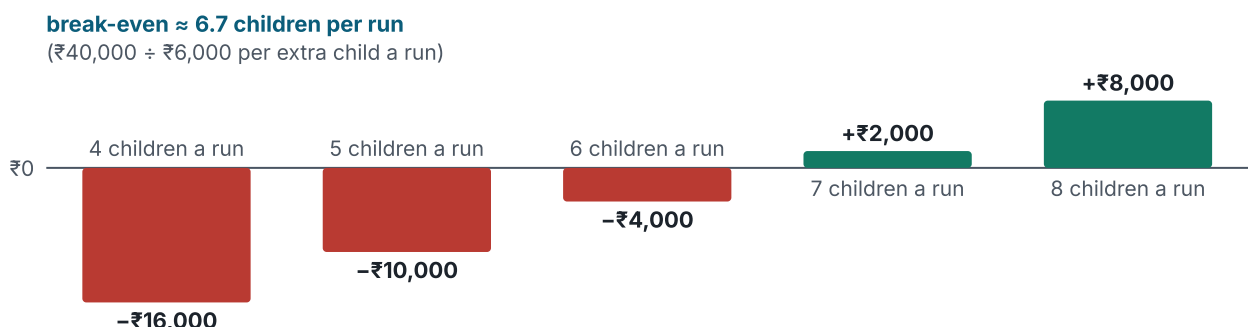


Figure 4.11 · Density decides the business. With 5 children a run the van loses ₹10,000 a month; with 8 it earns ₹8,000. Two school timings a morning make the same van and driver pay twice. *Illustrative: driver ₹18,000, attendant ₹7,000, van lease ₹9,500, fuel ₹5,500 a month.*

A ④ MVP

One school with two start times, three vans, 24 children. Ola staff build the routes by hand, so no new app yet. Parents get tracking through the existing app and a WhatsApp group with the driver. Price: ₹3,000 a month, first week free. Every driver is verified and every van meets the state's school-vehicle rules.

Why this move: people do by hand what software would do later. It costs weeks, not months, and it still tests real payment and real mornings. Safety is not cut, because it can't be tested later.

A ⑤ LEARN

Before starting, I'd write the pass mark. Of 600 van families told about it, do 5% (30) sign up and pay? Do 80% renew after month one? Is the van at the gate within 5 minutes of the bell on 95% of days? Are there at least 7 children a run? If price fails, test ₹2,500 with a smaller van. If density fails, shrink the radius to 3 km.

Why this move: a pass mark written in advance stops the team from calling any result a success.

A ⑥ SCALE AND MEASURE

Only then build seat booking and routing in the app, and sign school partnerships, since schools reach parents most cheaply. **Goal (North Star):** children delivered to the gate on time each school day. **Safety checks (guardrails):** safety incidents (target zero), seats over the cap, driver churn. **Trade-off:** dedicated drivers make it reliable but cost Ola the flexibility of on-demand supply. **Next time:** offer the afternoon return run to drivers who are otherwise idle then.

Why this move: the goal counts a promise kept to parents (on time, every day), which is what makes them renew.

HOW THE ANSWER HANGS TOGETHER

The goal (predictable revenue) led to a subscription. The van maths (break-even at about 7 children a run) exposed the two riskiest assumptions: price and density. The MVP (3 vans, 1 school, routes by hand) tests exactly those, with a pass mark set in advance. Safety is the one thing never cut, and it becomes the guardrail.

FRAMEWORK · MVP, BUILD-MEASURE-LEARN AND THE RISKIEST ASSUMPTION

1 · The situation. The candidate designed the full service and called it the MVP. Nothing in it would tell Ola, cheaply, whether parents will pay or whether vans will fill.

2 · An everyday example. Meena wants to start a home tiffin service. She could rent a kitchen, hire two delivery boys and print menus. Instead, she cooks for five neighbours for two weeks at ₹120 a meal, collects cash, and notes who orders again. If three of five reorder, she has learned more than any business plan could tell her.

Idea	Meena's tiffin service	Ola for schoolchildren
The big plan	Kitchen, staff, menus, an app	A new app section, a fleet, school tie-ups
Riskiest assumption	People pay ₹120 and order again	Parents pay ₹3,000 and renew; vans fill 7+ seats
The MVP	Cook for five neighbours	3 vans for 1 school, routes by hand
Measure	Reorders after week one	Sign-ups, renewals, on-time arrivals
Learn	Change the price or the menu	Change the price or the route radius

3 · The framework. An **MVP** (minimum viable product) is the smallest thing that tests your riskiest assumption with real users and real money. It is *not* version 1 with fewer features. The **build-measure-learn loop** (from Eric Ries's *The Lean Startup*) says: build the MVP, measure what users actually do, learn, and go round again, faster each time. To choose what to test, list every assumption and place each one by how little you know and how fatal it is if wrong (Figure 4.12). Test the top-right corner first. Common MVP types: **concierge** (people do by hand what software will do later), **Wizard of Oz** (it looks automatic to users, but people run it behind the scenes), and **fake door** (a button or landing page that measures interest before anything exists).

Using it, step by step (on this case):

1. Write every assumption as a sentence that could be false: "parents will pay ₹3,000 a month".
2. Place each on the map: how little we know, how fatal if wrong. Price and density land top-right.
3. Pick the cheapest test that uses real money and real behaviour: 3 vans, 1 school, first week free, then pay.
4. Write the pass mark before starting: 30 paying families, 80% renewal, 7 children a run.
5. Run one loop in about four weeks, then decide: go on, change price or radius, or stop.

COMMON SLIP · building a smaller app and calling it an MVP. It takes months and tests whether the screens work, while the real unknown is whether anyone pays.

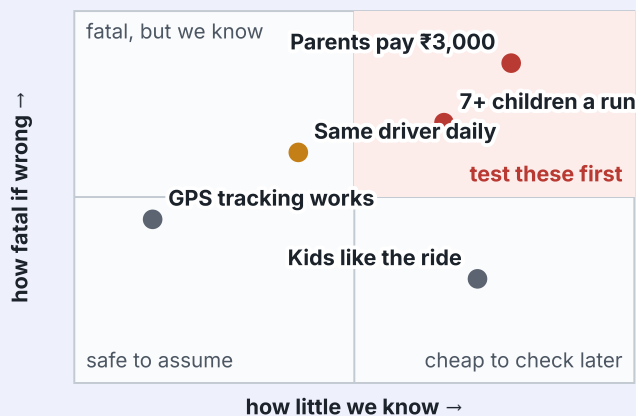


Figure 4.12 · Where to point the first test. Price and density are unknown and fatal; tracking is known; how children feel matters but can be checked later. *Author's placement.*

4 · When it misleads. A concierge MVP can succeed because of effort that won't scale. If the PM personally calls every parent each morning, parents love it, and the real service without those calls may fail, just as Meena's tiffin tastes great because Meena cooks, not the cook she will hire. Some things also can't be "minimum": a weaker background check is never a valid test for a children's service.

5 · In the interview. "Before designing the full service, I'll name the assumption that would kill it, and design the smallest real test of that one assumption."

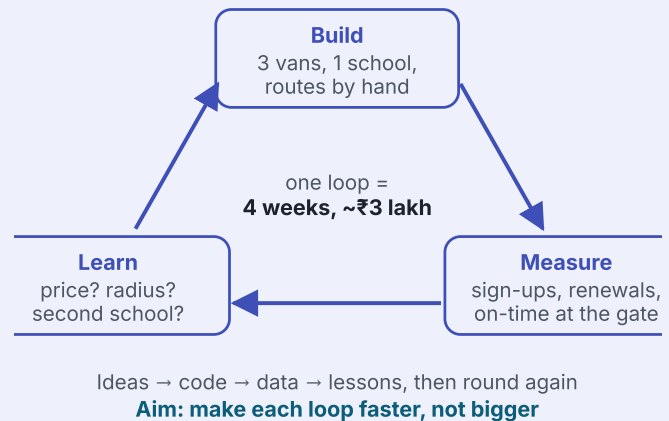


Figure 4.13 · One build-measure-learn loop. Each loop answers one question. Speed of learning, not the size of the build, is the goal. *After Ries, The Lean Startup (2011); numbers illustrative.*

CASE 4.2 · IMPROVE GOOGLE MAPS

Product improvement

Medium IIM B Sigma PM Casebook 2026–27, pp. 114–115

Try it first. Open Maps and look at the last five things you did in it. How many were "take me from A to B", and how many were "help me decide where to go"? Which kind would you want more of, if you ran Maps? Five minutes, out loud.

THE QUESTION

How would you improve Google Maps? (VMock)

- A** Let me start with a few clarifying questions. What do you mean when you say improve? Is the goal to enhance engagement, acquire more users, or to generate more revenue?
- Q** Let's go with engagement
- A** Sure, and do we have any specific platform in mind? iOS app, Android app, or website? Or is it across all platforms?
- Q** Around 80% of our users are on Android so let's start with that. We can later expand it to iOS and web.
- A** Sure. So first I'll start with listing down a few kinds of users for the app –
- 1. Urban Millennials:** These are young folks in 20-35 years age group who are tech savvy, have high end phones, and good connectivity. They mostly use google maps for same city travel to schools, offices, new spots to visit on weekends etc.
 - 2. Cab / Delivery Drivers:** The driver fleets for various cab services, e-commerce websites, food delivery platforms use Google Maps daily for their last mile connectivity.
 - 3. Tourists:** Tourists use google maps to find their way in a new city for finding vacation spots and restaurants. Among these, I would like to proceed with the Tourists segment as I feel they rely a lot on maps when in a new city. Urban millennials mostly use maps for same city travel and have most frequently used routes memorized. Delivery drivers, although are heavy users, only use the app to reach from point A to point B in the shortest possible time and Google Maps is currently serving that need very well.
- Q** That sounds great, please go ahead.
- A** The typical user journey through the app looks something like this -

The original shows a diagram here; it is redrawn as Figure 4.14, labelled "redrawn from the original".

A In this journey, I can think of the following pain points -

1. Trip Planning: It is difficult to narrow down the places to visit while travelling. We can find the list of “top 10 places” online but those are not curated to interests, location, and period of stay of the user. For example, if a person is really interested in history and has a really short stay, they would ideally like to visit the museums.
2. Authentic Experience: People usually look for the authentic local experience when travelling. This may include finding the best places to try the local food, or some local street shopping. One prominent issue is that they have to spend a lot of time researching the reviews and ratings of restaurants, and even local shopkeepers try to extort more money from foreigners.
3. Managing Bookings: Tourists need to buy tickets for museums, palaces, gardens, public transport, etc from different websites or stand in long queues at the counters. This can be very cumbersome.
4. Group Travel: Often tourists also rent cars to explore around the city. For a large group with multiple cars, it can get quite difficult to keep track of people in different cars.

Q Why don't you go ahead and tell me the solution for these?

A I can club the first 3 pain points and address those using a ‘Travel Planner’ feature. The 4th pain point I believe is faced by a small subset of travelers only and can be neglected for now. The Travel Planner will take the following inputs from the user -

- Location of the user: Can be entered manually (for pre-planning) or auto detected in Google Maps (for impromptu plans)
- Interests of the user: Explicitly asked through checkbox options.
- Duration of stay: Explicitly asked

Using these inputs the Travel Planner would create a personalized inventory for the user. It will include all tourist hotspots that align with the user’s interests – museums, theme parks, galleries, monuments, street shopping locations, restaurants etc. The users are obviously free to rearrange the places and add/drop suggestions from the generated list. Once the users start their journey, they will be able to navigate to the next destination in the itinerary. After reaching the destination, Google Maps can also display some additional information about the places such as some historical facts, trivia, and reviews from other people who have visited before. There will also be an option to pre purchase tickets (or make reservations) for these destinations directly from the app itself.

Q Interesting. Do you think there is any way to make money out of this as well?

A Currently all features on the app are free for end users so I don’t want to add any paid aspect for them. The ticketing aspect however can help in generating additional revenue. Maps can take a listing fee for allowing businesses to sell tickets / make reservations on their platform. Alternatively, they can also charge a commission from the overall transaction value of tickets booked through Maps.

Q Okay, and how would you measure the success of your feature after the launch?

A We could look at the number of users per month using the Travel Planner to generate an itinerary, the number of users who make bookings through maps itself. The revenue generated from commissions can also be noted.

Q Sounds good. Do you think this feature could impact any existing feature? What would the trade-offs be?

A The app has different categories like hotels, restaurants etc. listed below the search bar that people use to find nearby destinations. The click through rate on these categories may drop for people who are using the Travel Planner. However, this shouldn’t be a problem as we are still helping people find the places they are looking for.

Q That’s great, thank you.



Pain points the candidate names, placed on the stage where they bite:

1 Trip planning

4 Group travel

2 Authentic experience

3 Managing bookings

Redrawn from the original (IIM B casebook, p. 114); pain-point placement added.

Figure 4.14 • A tourist's journey through Maps, as the candidate drew it. Three of the four pain points bite before and during exploring, not while navigating, which is where Maps is already strong. Redrawn from the original, IIM B Sigma PM Casebook 2026–27, p. 114; pain-point placement is the author's.

WHAT WORKED • WHAT WAS MISSING

- ✓ **Asked the goal and the platform** before anything else, and got two useful constraints: engagement, Android first.
- ✓ **Chose a segment with a reason:** commuters know their routes, drivers are well served, tourists depend on Maps in a new city.
- ✓ **Mapped the journey** and placed pain points on it, then grouped three into one feature.
- ✓ **Made money without charging users:** a commission on bookings, and saw the risk to existing category chips.
- ✗ **Never defined engagement.** Tourists use Maps for a few days a year; “more engagement” needs a measurable meaning.
- ✗ **Didn't check what exists.** Maps already has Explore, saved lists and, since March 2026 in India, Ask Maps: a Gemini-powered way to ask “what should I do here for two days?”
- ✗ **Metrics are counts, not outcomes.** “Users who generate an itinerary” rises even if nobody follows it. No guardrail at all.
- ✗ **Commission can bend the plan.** If Maps earns on bookings, will the itinerary favour places that pay? Trust is Maps' most valuable asset.
- ✗ **No cheap test** before building a full planner.

Scoping kit check: Confirm the thing ✗ **which part of Maps** • Goal ✓ **engagement** • Who ✓ **tourists** • Today and the trigger ✗ **existing features, why now** • Edges ✓ **Android first**. Skipping “today” is why the answer designed a planner that overlaps with Ask Maps and lists; one question would have turned it into an improvement of what is there.

CHECK • “DELIVERY DRIVERS... GOOGLE MAPS IS CURRENTLY SERVING THAT NEED VERY WELL” **OVERSTATED**

Getting from A to B is well served. Finding the exact door in India is not: addresses are often descriptive (“behind the temple, second lane”) and pins are frequently wrong. Delivery and cab apps build their own address fixes, and in 2024 Google itself added landmark-based address descriptions in India. Heavy daily users with a hard last 100 metres are a strong segment, not a solved one.

STRONGER ANSWER



Figure 4.15 • An improvement answer built as a tree. One outcome at the top; every idea hangs from a user need. Author's summary.

A ① CLARIFY

By engagement, do we mean more days people open Maps, or Maps used for deciding where to go, not only for directions? And what should I take as given: Explore, saved lists and Ask Maps already exist. I'll assume India, Android first.

Q Planning and discovery, not just directions. Yes, those exist.

Why this move: now the answer improves what exists, and success means Maps being used to decide, which is where the money (local ads, bookings) is.

A ② OUTCOME

I'll keep tourists, including domestic ones: India records about 250 crore domestic tourist visits a year. One outcome: the share of trips outside a user's home city where they save or follow a plan in Maps. Illustrative: 10 crore users × 2 trips a year = 20 crore trips. Moving planned trips from 8% to 15% = 20 crore × 7% = 1.4 crore more trips planned in Maps.

Why this move: a single outcome makes every later idea comparable; the sizing shows the prize is worth a team.

A ③ OPPORTUNITIES

From user research I'd expect four needs, written in the tourist's words (Figure 4.16): "I don't know what's worth it in two days"; "is this a tourist trap, will I be overcharged?"; "queues and tickets everywhere"; "I can't read the menu or signs". The first two affect almost every short trip; I'd start there.

Why this move: needs come before features. The candidate went from pain to a "Travel Planner" in one step, and never compared it with other ways to meet the same need.

A ④ SOLUTIONS

For "what's worth it": a **Plan my day** card built from places I saved, open hours and travel time, and Ask Maps prompts for trips. For "tourist trap": a **Locals love this** badge, based on the share of reviews from people living within 20 km, and a price range from recent visitors. The badge is cheap to build and helps every tourist, even those who never plan.

Why this move: two or three solutions per need let you pick on cost and reach, instead of defending your first idea.

A ⑤ TEST CHEAPLY

A **fake door**: show a "Plan my day" button to 1% of users who search in a new city; count taps before building anything. And the badge on places in two cities, compared with two similar cities without it: do saves and visits rise?

Why this move: the riskiest assumption is that tourists want a ready plan at all. A button that measures taps answers that in a week.

A ⑥ MEASURE

Goal (North Star): share of trips with a plan saved and at least two stops visited. The HEART table below gives the supporting signals. **Safety checks (guardrails)**: share of plan stops that are paying partners, so bookings never bend the plan; complaints about closed or wrong places. **Trade-off**: bookings bring revenue but can cost trust. **Next time**: tackle tickets and queues through partners.

Why this move: counting visits, not plans created, rewards plans people actually use; the partner-share guardrail protects the thing Maps can't afford to lose.

HOW THE ANSWER HANGS TOGETHER

Defining engagement as "Maps used to decide" set one outcome: trips planned in Maps. The tree hung every idea from a tourist's need, the fake door tests the biggest assumption before building, and the guardrail on paid stops keeps the new revenue from eating the trust that makes Maps worth planning with.

FRAMEWORK · OPPORTUNITY SOLUTION TREE

1 · The situation. The candidate jumped from pain points to one feature. There was no way to see other solutions, or to tell which need mattered most.

2 · An everyday example. You want to lose 5 kg by March. You notice three reasons you haven't: late-night snacking, no time to exercise, four cups of sweet tea a day. For each, there are a few fixes: keep no snacks at home, walk during phone calls, cut sugar to half a spoon. You try one for a week and weigh yourself. That is a tree: one goal, the reasons, the fixes, the tests.

Level	Losing 5 kg	Google Maps for tourists
Outcome	5 kg lighter by March	15% of trips planned in Maps
Opportunities	Night snacking; no time to exercise; sweet tea	"What's worth it?"; "tourist trap?"; tickets; language
Solutions	No snacks at home; walk on calls; half sugar	Plan my day; Locals love this badge; price range
Experiments	One week without evening snacks	Fake-door button; badge in two cities

3 · The framework. Teresa Torres's **Opportunity Solution Tree** (from *Continuous Discovery Habits*, 2021) has four levels. The **outcome** is one measurable change the team owns. **Opportunities** are user needs, pains or wishes, found in interviews and written in the user's words. **Solutions** are ideas that could meet an opportunity, several per opportunity. **Experiments** are the cheapest ways to test whether a solution works. You work top-down, pick one opportunity at a time, and compare solutions side by side (Figure 4.16).

Using it, step by step (on this case):

1. Write one outcome with a number: trips planned in Maps, 8% to 15%.
2. List the opportunities from what tourists say, never as features: "I don't know what's worth it in two days".
3. Pick the one or two that most tourists feel on most trips.
4. Under each, write at least two solutions, and one that is cheap.
5. Test the riskiest solution with the cheapest experiment, then update the tree.

COMMON SLIP · writing a feature as an opportunity (“users need a travel planner”). A need never names the solution; if it does, you have skipped a level.

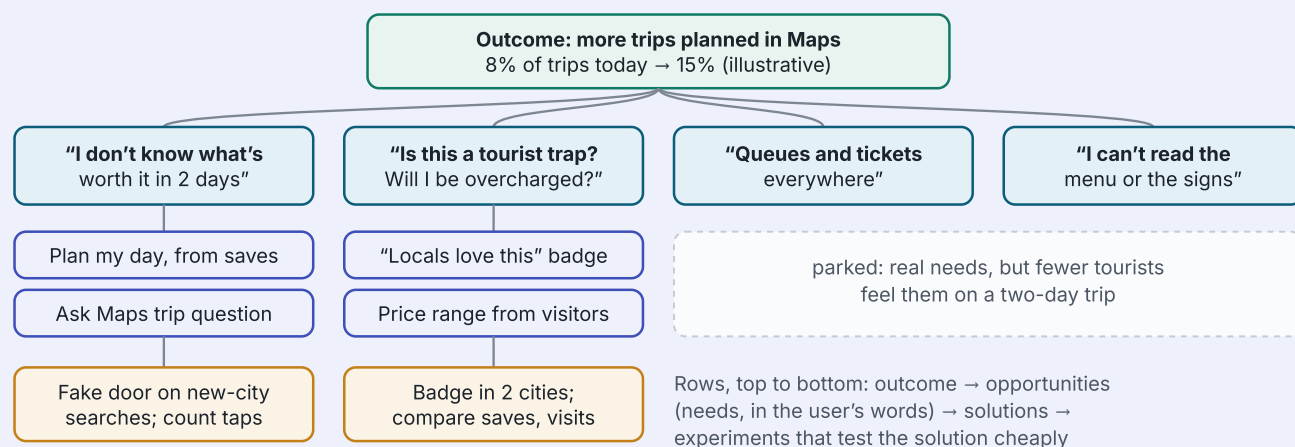


Figure 4.16 · An Opportunity Solution Tree for tourists in Maps. Two opportunities are worked on, two are parked on purpose. Each experiment tests a solution before anyone builds it. *Structure after Torres (2021); content and numbers illustrative.*

4 · When it misleads. A tree built in a meeting room is a list of guesses drawn nicely. If the opportunities didn't come from real tourists, the tree only makes your first idea look researched. It can also grow into a backlog of fifty boxes; prune it to what you will work on this quarter.

5 · In the interview. “I'll fix one outcome, list what tourists struggle with in their words, and only then compare two or three ideas for the biggest need.”

FRAMEWORK · HEART METRICS

1 · The situation. The candidate measured users who make itineraries and bookings. Both can rise while tourists find the plans useless.

2 · An everyday example. A restaurant adds a new biryani. The owner asks five questions: do diners say it's good (happiness)? Do regulars order it often (engagement)? Do new customers try it (adoption)? Do they order it again next month (retention)? Does it reach the table hot, in 15 minutes (task success)?

3 · The framework. Google's **HEART** framework (Rodden, Hutchinson and Fu, 2010) scores a feature on **Happiness, Engagement, Adoption, Retention and Task success**. For each letter you write a **goal**, a **signal** (what users do or say that shows it) and a **metric** (the number you track). Pick the two or three letters that matter for this feature; you rarely need all five.

Letter	The biryani (everyday)	Plan my day: goal	Signal → metric
Happiness	Diners say it's good	Tourists trust the plan	Rating after the day → share rating 4 or 5
Engagement	Regulars order it often	Plans guide the day	Stops visited from the plan → stops per trip day
Adoption	New customers try it	Tourists find it	First plan opened → % of trips with a plan opened
Retention	Ordered again next month	A trip habit	Used again next trip → % using it on 2 trips
Task success	Hot, on the table in 15 min	Works on the ground	Stops open and reachable → % visited as planned

Using it, step by step (on this case):

1. Name the feature: Plan my day.
2. For each letter, ask whether it matters here. All five do, but task success and retention matter most.
3. Write goal, then signal, then metric, in that order, so the metric follows from the goal.
4. Choose one as the headline (task success) and add a guardrail from outside HEART (paid-partner share).

COMMON SLIP · starting from the metrics you already log. Goals first; otherwise HEART just relabels your dashboard.

4 · When it misleads. HEART measures a feature, not the business. A plan feature can score well on all five while it quietly moves bookings to partners who pay the most. Add the business and trust guardrails yourself.

5 · In the interview. “For this feature, I care most about task success: did the plan actually work on the ground? I'd back it with retention on the next trip.”

Try it first. You are an auto driver at Majestic at 9 pm. A passenger wants to go to a far suburb, and you know you will drive back empty. What do you do, and what would change your mind? Then ask what an app could change.

THE QUESTION

We're facing a sudden spike in ride cancellations on the Uber app, and customers are getting poor service as a result. Walk me through how you'd find the root cause.

- A** To start, can you clarify what the app does here and which service the cancellations affect?
- Q** The app connects riders with drivers for transport. The issue is ride cancellations.
- A** Helpful. One clarification on the metric: are these rider-initiated or driver-initiated cancellations - or both? That single split will narrow the search a lot, because the causes are quite different.
- Q** Good question - they're mostly driver-initiated.
- A** Then I'll focus on driver behaviour. Over what period has this spike been seen?
- Q** Recent, mainly the last couple of weeks.
- A** Any geographic concentration?
- Q** More prevalent in metros, but not limited to one city.
- A** Which customers are most affected - new, existing, or both?
- Q** Both, but more complaints from frequent riders.
- A** Which platform - iOS, Android, both?
- Q** Both, so it isn't platform-specific
- A** Any change to how the cancellation metric is calculated or reported?
- Q** No change to calculation or reporting.
- A** Now the journey: a rider opens the app, enters a destination, requests a ride, and waits for a driver to accept; they're then shown the driver's details, price and ETA. Since the cancellations are driver-side, the relevant question is what changed for drivers. Any recent release affecting the driver app or experience?
- Q** Yes - we rolled out an update letting drivers see the drop-off location before they accept a ride.
- A** At which point in the flow are the cancellations happening?
- Q** Drivers usually cancel after accepting but before reaching the pickup.
- A** Any technical glitches reported with the new feature?
- Q** No technical glitches.
- A** Any other recent change - pricing, driver incentives, service quality, new offerings that could shift driver behaviour?
- Q** No, the main change is drivers seeing the drop-off location.
- A** And externally - new regulations, environmental events, competitor promotions, or PR/social sentiment affecting drivers?
- Q** None that we're aware of.
- A** Then the evidence points clearly to the drop-off-visibility feature. By accepting and then seeing or by seeing then cancelling just after accept, drivers are declining trips with undesirable drop-offs (long dead-heads back, low-demand areas, traffic), which matches the after-accept, before-pickup timing and the driver complaints. I'd address it by rebalancing driver preferences with rider needs: incentives or surge for less-desirable drop-offs, a cap or penalty on post-acceptance cancellations, and possibly showing drop-off only after a commitment window. I'd validate by correlating cancellation rate with drop-off characteristics and confirming the spike tracks the feature's rollout.
- Q** That makes sense. Thank you for walking through it so thoroughly.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Split driver from rider cancellations first**, and said why: the causes differ completely.
- ✗ **Never asked how big.** From 8% to 14%, or from 8% to 9%? And did completed trips actually fall, or were rides quickly reassigned?

- ✓ **A near-complete checklist:** timing, city, rider type, platform, metric definition, releases, point in the flow, glitches, other changes, outside events.
- ✓ **Found where in the flow** it happens (after accepting, before pickup), which pointed straight at the release.
- ✓ **Proposed a way to confirm it:** cancellation rate against drop-off type, and the timing of the rollout.
- ✗ **Didn't ask why drivers accept and then cancel**, instead of simply declining. The usual answer: an acceptance-rate target in driver incentives. That is the real bug.
- ✗ **The release revealed bad trips; it didn't create them.** Before it, drivers found out at pickup and cancelled or rushed the rider. The cause is the economics of trips that end far away.
- ✗ **Two of the three fixes fight drivers.** Penalties and hiding the drop-off push drivers who run two apps towards the rival.

Scoping kit check: Confirm the thing ✓ · Goal ✗ · Who ✓ **driver-side, frequent riders** · Today and the trigger ✓ **last two weeks** · Edges ✓ **metros, both platforms**. RCA add-ons: exact definition ✓ · size vs baseline ✗ · where it shows ✓ · recent changes ✓ · is the data trustworthy ✓. One of the best openings in the casebook. The one gap, size, is also what would have shown whether riders were really worse off.

CHECK · "A CAP OR PENALTY ON POST-ACCEPTANCE CANCELLATIONS" · DIRECTIONALLY RIGHT

India's 2025 aggregator guidelines allow a penalty of 10% of the fare, up to ₹100, on a driver or rider who cancels without a valid reason, so a penalty is legal. But a driver facing an hour of empty driving back will still cancel, or will simply switch to the other app for the evening. Penalties work only after the trip itself is worth taking.

STRONGER ANSWER



Figure 4.17 · The RCA sequence, with a fourth step that reads the driver's maths. In a two-sided market, "why" usually means "why is this rational for one side?" *Author's summary.*

A ① CLARIFY

Which cancellations, driver or rider? How big is the rise, from what to what? And the outcome that matters: did completed trips per request fall, or were cancelled rides quickly given to another driver?

Q Driver-side, after accepting: 8% to 14% of accepted rides in two weeks. Completed trips per request fell from 78% to 73%.

Why this move: a 5-point fall in completed trips per request is real harm to riders, so this is worth fixing now, not a reporting quirk.

A ② RULE OUT

Same checks as the candidate, done quickly: no change in how the metric is counted, no outage, no weather or strike, no competitor offer. The one change is the release that shows drivers the drop-off before they accept.

Why this move: ruling out the boring causes first (data, outside events) protects you from blaming a release for a festival or a rainstorm.

A ③ FIND WHERE

I'd split cancellations by where the trip ends. Illustrative: inside the city core they barely moved, 8% to 9%. To the airport, 10% to 24%. To the city's edge, 11% to 31% (Figure 4.18). Most cancel within a minute of accepting.

Why this move: one split turns "drivers are cancelling" into "drivers are cancelling one kind of trip", which tells you the cause.

Driver cancellations after accepting, % of accepted rides, by where the trip ends (illustrative)

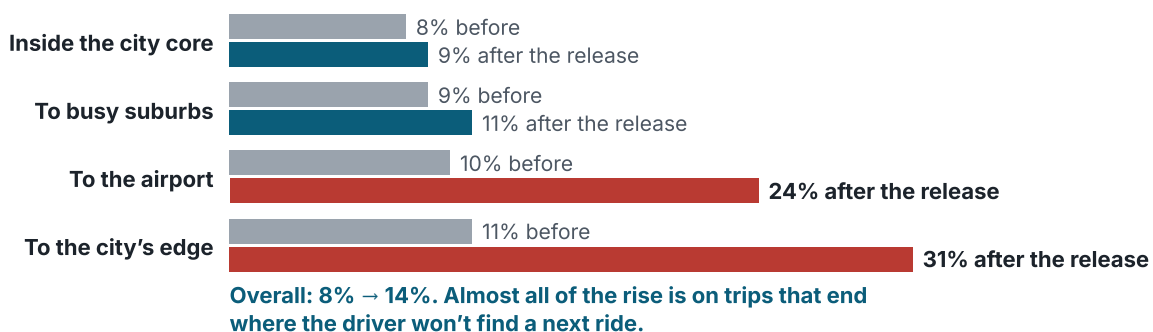


Figure 4.18 · Where the cancellations come from. The rise sits almost entirely on trips that end where drivers won't find a next fare. *Illustrative numbers consistent with the case (a two-week rise after a driver-app release).*

A ④ WHY: THE DRIVER'S MATHS

A city trip nets the driver about ₹117 for 55 minutes: ₹128 an hour. An edge trip: ₹600 fare – ₹120 commission – 53 km of fuel at ₹5 (₹265, because the car comes back empty) – ₹48 EMI = ₹167, for 70 minutes out and 60 back. ₹167 ÷ 130 minutes × 60 = ₹77 an hour, 40% less. So why accept first? I'd ask about incentives.

Q Weekly bonuses need a 90% acceptance rate.

Why this move: now both halves make sense. The trip pays badly, and the bonus punishes declining it, so drivers accept, look, and cancel. Nobody is misbehaving; the incentives are.

A ⑤ FIX

Three changes. **Pay for the return:** a return premium on trips ending in low-demand zones, split between rider and platform. **Match by direction:** offer edge trips first to drivers heading that way, such as those ending their shift near there. **Fix the bonus:** count cancellations after accepting, not declines, so an honest “no” costs the driver nothing. Cost, illustrative: 8% of trips × ₹100 premium, half paid by the rider = ₹4 a trip on average for the platform.

Why this move: each fix removes a reason to cancel rather than punishing the cancel. Hiding the drop-off again would only move the problem to the pickup.

A ⑥ MEASURE

Goal (North Star): completed trips per request, back to 78% or better. **Inputs:** cancellations after accepting on edge trips; time to match edge trips. **Safety checks (guardrails):** driver earnings per online hour; rider complaints about the premium; ETA in the city core. **Trade-off:** letting drivers decline freely makes edge trips slower to match, but a quick “no” wastes less of the rider's time than a “yes” that is cancelled. **Next time:** read every driver-facing release by its effect on earnings per hour.

Why this move: the goal measures what riders feel (did I get a ride?), while the driver-earnings guardrail makes sure the fix doesn't work by squeezing drivers.

HOW THE ANSWER HANGS TOGETHER

Sizing (8% to 14%, completed trips down 5 points) showed real harm. The split by drop-off found *which* trips, the driver's maths (₹77 an hour against ₹128) found *why*, and the bonus rule explained *why after accepting*. Each fix removes one of those reasons, and the goal and guardrail check that riders get rides without drivers paying for it.

FRAMEWORK · SUPPLY-DEMAND BALANCE AND SURGE PRICING

1 · The situation. Nobody in the case mentioned price, yet drivers were voting on price with every cancellation. A ride that pays too little for the time it takes is a market out of balance, one trip at a time.

2 · An everyday example. Bengaluru's railway station at 1 am. A late train arrives and 300 people want autos; 60 autos are waiting. Drivers quote double the meter. At that price, a few more drivers come out from home nearby, and some passengers decide to wait for the first bus. The high price is ugly, but it gets more people home than a fixed price with a two-hour queue.

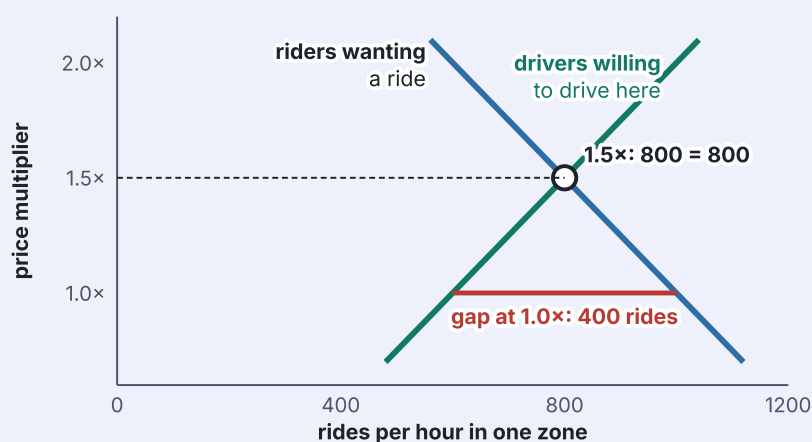
Idea	The station at 1 am	Uber on a rainy Friday
Demand	Passengers off the late train	People leaving offices in the rain
Supply	Autos at the stand	Drivers nearby and free (not stuck with a rider)
Price signal	“Double meter”	1.8× surge on the fare
Supply responds	Drivers come out from home	Drivers drive in or log on
Demand responds	Some wait for the bus	Some take the metro or wait an hour
Bad trips	“Not going that side”	Cancels on edge and airport trips

3 · The framework. At any price, some number of riders want rides and some number of drivers are willing to drive. When the price is too low, riders outnumber drivers: waits grow, drivers pick the best trips and cancel the rest. **Surge pricing** (a temporary multiplier on the fare) raises the price until the two meet. Worked example (Figure 4.19): at 1.0×, 1,000 riders want a ride and 600 drivers are available, a gap of 400. At 1.5×, demand falls to 800 and supply rises to 800. The gap closes. The same logic works for a single trip: a trip that pays less per hour than the driver's alternative will be declined or cancelled, whatever the rules say.

Using it, step by step (on this case):

1. Name the two sides and the unit: riders and drivers, rides per hour, in one zone.
2. Ask which side is short, where and when: drivers willing to take edge trips at night.
3. Find the driver's alternative: ₹128 an hour on city trips against ₹77 on an edge trip.
4. Change the price of the short side's bad option (a return premium) or change who gets it (drivers heading that way).
5. Check both reactions: did more drivers accept, and did riders still book at the new price?

COMMON SLIP · treating surge only as a way to earn more. Its main job is to move drivers to where riders are; if it doesn't change supply within 10–15 minutes, it is only a tax on riders.



At base price (1.0x)

1,000 want a ride, 600 drivers nearby: 400 wait, cancel or leave. Drivers pick the best trips and cancel the rest.

At 1.5x surge

200 riders take the metro or wait; 200 more drivers drive in or log on. Nobody waits long.

Capped at 2x base (2025 rules):

if the gap needs more, it stays.

Figure 4.19 · Why a higher price can shorten waits. At the base price the red gap is riders who wait, cancel or give up. Surge closes it from both ends: fewer riders, more drivers. *Illustrative curves; the 2x cap is from India's Motor Vehicle Aggregator Guidelines, July 2025.*

4 · When it misleads. In an emergency, such as a flood or an attack, supply can't respond and high prices look like exploitation. Most apps now cap or switch off surge in declared emergencies. Drivers also learn to game it: there have been reports of drivers logging off together near airports to trigger surge. And a cap (2x under India's rules) can stop the price from ever closing a very large gap, so waits stay long however well the algorithm works.

5 · In the interview. "Cancellations are drivers telling us a trip is priced below what their time is worth. I'll find which trips, and fix the price or the matching for those trips."

CASE 4.4 · MAKEMYTRIP ENTERS DUBAI

Go-to-market

Medium

IIM B Sigma PM Casebook 2026–27, pp. 151–152

Try it first. A friend who runs a successful dosa restaurant in Chennai wants to open in Dubai. What three questions would you ask before saying yes? Now ask the same three about MakeMyTrip.

THE QUESTION

MMT, is considering entering the Dubai market. What should their market entry strategy be?

- A** Before starting with the strategy, I have some initial clarifying questions. By MMT, are we referring to the online travel platform, MakeMyTrip.
- Q** Yes !
- A** First, I would like to evaluate the desirability of entering the Dubai market. So, what are MMT's specific objectives and expectations for entering the Dubai market? Do we have any specific metrics like revenue or profit targets?
- Q** The firm's main objective is to maximize its revenue and ROI over the next five years. Dubai's travel market is rapidly growing, driven by increasing tourism and luxury travel. There is no specific revenue target, but they aim for significant growth.
- A** That sounds promising. What about the competitive landscape? Who are the key players, and what are their strengths and weaknesses?
- Q** The market is competitive with established players like Booking.com and local agencies.
- A** Are there any desirable customer demographics that I should focus on (e.g., luxury travel, budget travel, business travel) that MMT is targeting?
- Q** MMT should target a mix of high-income residents and tourists looking for comprehensive travel services.
- A** Understood. To build a comprehensive strategy, I would want to evaluate the business from 3 perspectives:
 1. Economic Analysis – To determine market size and potential market share
 2. Internal Capabilities – To assess MMT's feasibility in the market.
 3. **Execution Plan:** To outline the steps for market entry

I would want to start with economic analysis first. I would like to assume that the fixed costs are essentially zero because of an already established technical platform . So, the initial costs include marketing, partner-

ships, and regulatory compliance. Do let me know if my assumption is fine and are there any estimated costs that I should think of?

Q The cost drivers are fine. Also, the expected revenue is substantial, with a forecast payback period of 3-4 years and a strong ROI.

A Understood. Given these investments, what are the main risks we should be aware of when entering the Dubai market?

Q What do you think could be some of the potential risks?

A 1. Well, one potential risk could be regulatory changes. Given that Dubai's regulatory environment can be quite dynamic, changes in laws or regulations could impact our operations or increase our compliance costs.

2. Another risk could be intense competition. Considering that Dubai is an attractive market, there are likely several well-established local and international players. Competing against them might require significant effort and resources.

3. Besides regulatory changes and competition, there could also be risks related to cultural differences and customer preferences. MMT might need to invest in understanding the local market better to tailor its offerings effectively.

Q The risks are relevant. Proceed next

A Given the promising revenue prospects and manageable risks, the market entry seems economically viable. Now, I'll like to evaluate the feasibility. Next, we need to assess if MMT has the internal capabilities to enter the market successfully. Are MMT's current product offerings suitable for the Dubai market?

Q Yes, MMT's offerings, are well-suited for Dubai. However, they might need some localization.

A Localization is indeed crucial. Could you specify what kind of localization might be required? For instance, would we need to customize the language, payment options, or perhaps the types of travel packages offered? I could think of customizing the language to Arabic, offering region-specific payment options like local credit cards and digital wallets, and tailoring travel packages to include popular local destinations and experiences would be necessary.

Q All of those are relevant

A What about the technological infrastructure? Does MMT's current platform

Q The platform is robust but will need some adjustments to integrate local payment gateways and to handle multi-language support efficiently.

A It's good to know the platform is strong. Moving on to organizational structure, do we have the right teams in place to manage operations in Dubai? For example, do we need a local office with a dedicated team for marketing, customer service, and partnerships?

Q Yes, MMT can think of a local team, but you can skip this aspect for now

A Finally, we need to develop an execution plan. How should MMT enter the market? Should we expand organically, enter a joint venture, partner with local firms, or acquire a competitor?

Q What would you recommend?

A I recommend a partnership strategy. By collaborating with local travel agencies, hotels, and airlines, MMT can leverage their market knowledge and networks. Additionally, launching a targeted digital marketing campaign can help build brand awareness and attract customers. Developing localized travel packages will also cater to the preferences of Dubai's residents and tourists.

Q What timeline would you propose for this market entry?

A It can be divided into 3 phase:

1. Short-Term (0-6 months): Conduct market research, establish partnerships, and adapt the platform for local use.
2. Mid-Term (6-12 months): Launch marketing campaigns, start offering localized travel packages, and monitor market response.
3. Long-Term (1-2 years): Expand service offerings, optimize operations based on feedback, and aim for market share growth.

Q That sounds comprehensive. What are the next steps?

A The next steps are:

1. **Establish Local Partnerships:** Engage with local travel agencies, hotels, and airlines.
2. **Digital Marketing Campaign:** Initiate targeted campaigns to build brand presence.
3. **Regulatory Compliance:** Ensure compliance with local regulations and adapt services to meet cultural norms.

Q That was good. Thanks for your time

WHAT WORKED · WHAT WAS MISSING

- ✓ **Asked the objective** and got a usable one: revenue and return over five years.
- ✓ **A clear three-part structure:** is it worth it, can MMT do it, how to enter.
- ✓ **Concrete localisation:** Arabic, local cards and wallets, local packages.
- ✓ **Compared entry modes** (organic, joint venture, partnership, acquisition) and gave a phased plan.
- ✗ **Never asked what exists today.** MakeMyTrip has run a UAE site and app for residents since about 2014, with Arabic support, a Dubai office and Umrah packages. The real question is how to grow it.
- ✗ **No sizing.** “Expected revenue is substantial” was accepted without a number, so there was nothing to test the plan against.
- ✗ **Tourists book at home.** A visitor from London books Dubai on a UK site before flying. A Dubai business mainly reaches the 39 lakh residents.
- ✗ **No right to win.** Why would a resident leave Booking.com or Almosafer? MMT’s real edge is the India corridor: over half of the UAE’s 44 lakh Indians live in Dubai.
- ✗ **No success measure or stop rule** for a five-year bet.

Scoping kit check: Confirm the thing ✓ · Goal ✓ **revenue and ROI, 5 years** · Who ✓ **residents and tourists** · Today and the trigger ✗ · Edges ✗ **budget**. GTM add-on “standalone or add-on: existing assets, right to win” ✗. Skipping “today” cost the most: the plan re-invents a business MMT already runs.

CHECK · “THE FIXED COSTS ARE ESSENTIALLY ZERO BECAUSE OF AN ALREADY ESTABLISHED TECHNICAL PLATFORM” OVERSTATED

The code is reusable; the rest is not. UAE payment methods, VAT, licences, Arabic and Malayalam support, and airline and hotel contracts in a new market all cost money. The biggest cost is buying customers against rivals who already spend heavily on ads: in online travel, customer acquisition, not technology, decides the payback.

STRONGER ANSWER



Figure 4.20 · A market-entry answer that sizes before it plans. Moves ② to ④ decide whether to enter at all; ⑤ decides how. *Author's summary.*

A ① CLARIFY

MakeMyTrip already has a UAE site for residents. Is this about growing that, or something new, such as selling Dubai tours to every visitor? And what budget and horizon: five years, as you said?

Q Grow the UAE business, Dubai first. Five years. Budget is open if the case is strong.

Why this move: “grow what exists” changes the question from “can we enter?” to “where can we win?”, and it lets us use data MMT already has.

A ② SIZE IT

Illustrative, per year (Figure 4.21). **TAM:** all travel booked by Dubai’s 39 lakh residents: $39 \text{ lakh} \times 1.5 \text{ trips} \times ₹50,000 \approx ₹29,000 \text{ Cr}$. **SAM:** trips where MMT has an edge, by Indian-origin residents: $22 \text{ lakh} \times 1.2 \text{ trips} \times ₹45,000 \approx ₹12,000 \text{ Cr}$; online, 60%: ₹7,200 Cr. **SOM:** 15% of that by year five $\approx ₹1,080 \text{ Cr}$ of bookings, about ₹110 Cr of revenue at a 10% take.

Why this move: ₹110 Cr is about 1% of MMT’s revenue. Knowing that early tells you this is worth doing only as the start of something bigger.

A ③ FORCES

For a general travel app in Dubai, four of five forces are strong (Figure 4.22): global rivals discount hard, airlines sell direct, and customers compare prices in seconds. In the India corridor it looks different: Booking.com has fewer Kerala homestays and Indian holiday packages, and Indian families trust a brand they used at home.

Why this move: the forces don’t say “enter” or “don’t”; they say where the market is winnable.

A ④ RIGHT TO WIN

MMT has what rivals lack for this group: India's hotel and holiday inventory, support in Hindi and Malayalam, and a brand most Indian residents already know. Its costs to acquire them are lower, because many already have the app from India.

Why this move: a right to win is an advantage rivals can't copy quickly. Without one, a market is only attractive to whoever already leads it.

A ⑤ ENTRY: WEDGE AND PHASES

0–6 months: “going home” for UAE Indians: flights to Kochi, Mumbai and Delhi with baggage clearly shown, festival-season fare alerts (Onam, Diwali, Eid), UAE cards, Apple Pay and pay-later. **6–18 months:** India holidays and family packages, where the margin is. **18 months on:** the same offer for Indians in Saudi Arabia, Qatar and Oman, about 90 lakh across the Gulf. Only then add general Dubai staycations.

Why this move: a wedge (a narrow first offer where you are clearly better) wins a loyal base cheaply; the phases then widen it where margins and reach grow.

A ⑥ MEASURE, WITH A STOP RULE

Goal (North Star): gross bookings from repeat UAE customers. **Inputs:** cost to acquire a customer by channel; share of flight bookings that add a hotel or package. **Safety checks (guardrails):** contribution margin per booking, so share is not bought with discounts; refund complaints. **Stop rule:** if acquisition cost doesn't pay back within 18 months by month 18, stop spending beyond the corridor. **Risk:** MMT said in August 2026 that international travel was subdued by the West Asia conflict, so plan a slower case too.

Why this move: a stop rule written in advance protects a five-year bet from sunk-cost thinking; the margin guardrail stops “growth” that is only discounting.

HOW THE ANSWER HANGS TOGETHER

Asking what exists turned “enter Dubai” into “grow a small UAE business”. Sizing showed the prize is small unless it grows across the Gulf. The five forces showed the general market is hard but the India corridor is winnable, and MMT's assets give it a right to win there. So the entry is a narrow wedge (going home), widened in phases, judged on repeat bookings, with a margin guardrail and a stop rule.

FRAMEWORK · TAM, SAM AND SOM

1 · The situation. The candidate and the interviewer agreed revenue would be “substantial” without a number. Nobody could say whether Dubai was worth one team or twenty.

2 · An everyday example. You plan a biryani kitchen in Koramangala. All the biryani Bengaluru orders in a year is huge, but you can deliver only within 5 km, and only at ₹250 a plate or more. Of that, with three famous biryani brands nearby, you might win 5% in three years. The last number is the one your business plan should use.

Level	Biryani kitchen in Koramangala	MakeMyTrip in Dubai
TAM	All biryani ordered in Bengaluru	All travel booked by Dubai residents, ≈ ₹29,000 Cr
SAM	Orders within 5 km, at your price	Trips by Indian-origin residents, online, ≈ ₹7,200 Cr
SOM	The 5% you can win from the famous brands	15% of that by year five, ≈ ₹1,080 Cr of bookings

3 · The framework. **TAM** (total addressable market) is all the spending on the need. **SAM** (serviceable available market) is the part your product, price and reach can actually serve. **SOM** (serviceable obtainable market) is the share you can realistically win in a stated time, given competitors. Build each level by multiplication you can defend (people × trips × spend), and state the time frame.

Using it, step by step (on this case):

1. Define the need and the buyer: travel booked by people who live in Dubai.
2. TAM from the bottom up: residents × trips a year × spend a trip.
3. SAM: keep only what you can serve well: Indian-origin residents, booked online.
4. SOM: pick a share you can defend from rivals' positions and your edge, for a year you name (15% by year five).
5. Turn SOM into revenue (× take rate) and compare it with the cost of winning it.

COMMON SLIP · “if we get just 1% of a ₹29,000 Cr market...”. A share of the total means nothing unless you can say who those customers are and why they pick you.

TAM ₹29,000 Cr

All travel booked by Dubai's 39 lakh residents
 $39 \text{ lakh} \times 1.5 \text{ trips} \times ₹50,000$

SAM ₹12,000 Cr

Trips where MMT has an edge: Indian-origin residents
 $22 \text{ lakh} \times 1.2 \text{ trips} \times ₹45,000$

Online ₹7,200 Cr

... of which booked online
 $₹12,000 \text{ Cr} \times 60\%$

SOM ₹1,080 Cr

What MMT can win by year 5
 $₹7,200 \text{ Cr} \times 15\% \text{ share}$

Year-5 revenue $\approx ₹1,080 \text{ Cr} \times \sim 10\% \text{ take} \approx ₹110 \text{ Cr}$ a year: about 1% of MMT's FY26 revenue ($\sim ₹9,000 \text{ Cr}$).

The prize grows only if Dubai becomes the base for all 90 lakh Indians across the Gulf.

Figure 4.21 · From ₹29,000 Cr to ₹110 Cr. Each step removes spending MMT can't serve or can't win. The final number is small for MMT, which is itself the insight: Dubai is a beachhead for the Gulf, not a prize on its own. *Illustrative multipliers; residents from the Dubai Statistics Center (2025), Indians in the UAE from the Indian embassy (2025).*

4 · When it misleads. Top-down numbers can be made as large as the deck needs: define the market as “all Gulf travel” and TAM grows tenfold with no change in the business. And SOM often hides the real assumption. The 15% share is a guess until you know what it costs to win each customer.

5 · In the interview. “Let me size it three ways: all the spending, what we can serve, and what we can win in five years. The last one decides the plan.”

FRAMEWORK · PORTER'S FIVE FORCES

1 · The situation. The candidate listed risks (rules, competition, culture) but couldn't say whether Dubai's travel market is one where anyone makes money.

2 · An everyday example. A chai stall outside a college gate. Is it a good business? It depends on five things: how many other stalls are on the same pavement; how easily a new one can open; whether students can just buy a café coffee instead; whether the milk supplier can raise prices; and whether students can walk ten metres to a cheaper stall.

Force	Chai stall at the college gate	Online travel in Dubai
Rivalry	Three stalls on the pavement	Booking.com, Expedia, Agoda, Almosafer, Wego
New entrants	A new stall needs only a cart	An app is cheap; airline and hotel deals are not
Substitutes	Café coffee, cold drinks	Airline apps, offline agents in Karama
Suppliers	The one milk vendor nearby	Airlines (strong), hotels (weaker)
Buyers	Students compare prices across stalls	Residents compare fares in seconds

3 · The framework. Michael Porter's **five forces** (1979) explain how much profit an industry leaves for the companies in it. The stronger each force, the more profit leaks away: to rivals through price wars, to suppliers through high costs, to buyers through discounts. Rate each force high, medium or low, with one line of evidence, then say what the ratings mean for *your* move (Figure 4.22).

Using it, step by step (on this case):

1. Define the industry narrowly: online travel booking for Dubai residents.
2. Rate each force with one piece of evidence: airlines sell direct, so supplier power is high.
3. Read the pattern: four high forces means thin margins for a general OTA.
4. Re-rate a niche: for the India corridor, rivalry and buyer power fall.
5. Choose the niche where the forces are weakest and your assets are strongest.

COMMON SLIP · listing the five forces as a ritual and stopping. The value is in the verdict: “general OTA, unattractive; India corridor, attractive”.

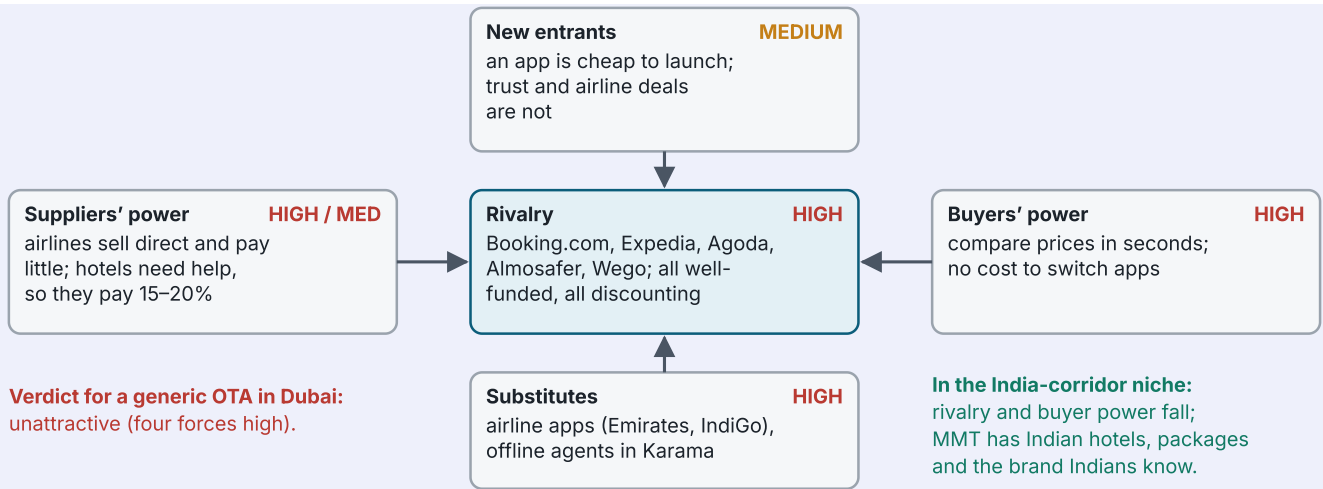


Figure 4.22 · The five forces for online travel in Dubai. Four forces are high for a general travel app, so profit leaks to rivals, airlines and price-comparing customers. The India-corridor niche changes two ratings. *Author's ratings, 2026; competitor list from public app stores and company sites.*

4 · When it misleads. The forces describe an industry as a snapshot, and they miss platforms and partners that help you. Google's search ads, for instance, are both a "supplier" of customers to OTAs and a rival. Use the forces to find where profit leaks, not to predict who wins.

5 · In the interview. "Before choosing how to enter, I'll check where profit leaks in this market. If it leaks everywhere, I'll look for a niche where it doesn't."

CHECK · "A SWOT ANALYSIS IS THE WAY TO STRUCTURE MARKET ENTRY" DOESN'T HOLD UP

SWOT (strengths, weaknesses, opportunities, threats) produces four lists with no weights and no decision. "Strong brand" sits next to "Arabic support needed" as if they mattered equally. And a strength means nothing without saying *in which market*: MMT's India inventory is a strength for Indian residents and irrelevant for European tourists. Sizing, five forces and a right to win do SWOT's job and end in a decision; use them instead.

MOBILITY, MAPS & TRAVEL · GUESSTIMATES

Guesstimates: sizing a fleet

Mobility guesstimates usually ask about supply: how many drivers, bikes, buses or servers. The trap is to count people instead of capacity. A fleet is sized for its busiest hour, and one vehicle can serve only so many trips in that hour. The six cards still do the work.

1

Pin down the unit

Drivers registered, active this month, or online right now? Cars only, or autos and bikes too?

2

Pick the approach

For fleets: supply from demand. Trips at the peak ÷ trips one vehicle does in that hour.

3

Draw the tree

People × share riding × trips → peak-hour trips → ÷ trips per vehicle per hour → ÷ share online at peak.

4

Round numbers, one reason each

A trip cycle includes the empty drive to the pickup, not just the ride.

5

Check it another way

Trips per driver per day, or trips per resident per day. Both have sensible ranges.

6

Give a range, and what moves it

Usually the share of people who ride, or the peak share of the day.

KEY IDEA · A FLEET IS SIZED FOR ITS BUSIEST HOUR

Fleet = peak-hour trips ÷ trips one vehicle completes in an hour, plus spares. If a quarter of a day's rides fall in two morning hours, sizing on the daily average leaves riders stranded at 9 am and vehicles idle at 3 pm. Every mobility business, and every server farm, is a peak problem first.

Try it first. How many times did you or your family take an app cab last month? Would the answer be similar for most people in a big Indian city? Build the number of drivers from that, for five minutes.

THE QUESTION

Can you quick estimate on how many uber drivers are there in Bangalore? (Samsung R&D)

A Sure. I'm assuming we are estimating the total number of uber drivers that serve Bangalore in a day. Is that correct?

Q Yes you are right please proceed.

A I'll calculate the number of drivers during the peak demand hours, let's say office hours. People are coming back from work on a Friday night. At a high level, we need to determine how many rides are demanded per peak hour and how many rides a driver can complete within that hour. We can estimate the number of drivers needed by dividing the total demand by the rides per driver per hour.

$$\text{Uber Drivers} = (\text{Drivers / Ride/Hours}) / (\text{Rides/Hour})$$

Q Sounds about right.

A The total population of Bangalore is close to 1.4 Cr. More than 50% of the population is between 20 and 60.

People in the age group of 20-60 years = $50\% \times 1.4\text{Cr.} = 70\text{L}$

Out of the assuming 60% are working professional = $60\% \times 70\text{L} = 42\text{L}$

Q Sounds like a fair estimate, go ahead.

A Not every one of them will use the cab assuming the following number-

1. No transport — 5%

2. Bus — 30%

3. Personal Vehicles — 20%

4. Cabs — $40\% = 40\% \times (42\text{L}) = 17\text{L}$ approx.

5. Others — 5%

There are other cab services like Ola, Rapido, and Meru since Uber and Ola are the top 2 players.

Assuming Uber Market share = 40%

Uber Customer = $40\% \times 17\text{L} = 6.8\text{L}$

Some people might be sharing cabs with friends and colleagues. On an avg. we can assume 2 people per ride.

No. of rides = $6.8\text{L} / 2 = 3.4\text{L}$ approx.

Q Sounds good.

A Let's say it takes 30 min to complete a ride. So, a rider makes 2 trips in an hour. Therefore, the number of Uber drivers = $3.4\text{L} / 2 = 1.7\text{L}$ drivers.

Q Seems practical. That would be it. Thanks!

WHAT WORKED · WHAT WAS MISSING

✓ **Pinned down the unit** (drivers serving the city in a day) and confirmed it.

✓ **Chose supply from demand at the peak**, the right approach for a fleet.

✓ **Stated the equation first**, then filled in numbers, and included shared rides.

✗ **40% of working people commute by cab** is far too high for any Indian city; buses, metro and two-wheelers carry most work trips.

✗ **Everyone rides in the same hour.** 3.4 lakh rides in one peak hour means all Uber commuters leave work at once.

✗ **A 30-minute ride is not a 30-minute cycle.** The empty drive to the pickup and waiting add 10–15 minutes.

✗ **No check.** 1.7 lakh drivers doing even 8 trips a day is 13.6 lakh trips, one for every 10 residents daily.

Guesstimate kit check: 1 Unit ✓ · 2 Approach ✓ **supply from peak demand** · 3 Tree ✓ · 4 Numbers with reasons ✗ **the cab share had no reason** · 5 Check ✗ · 6 Range ✗. Card 5 would have caught the error in seconds: the answer implies more cab trips than Bengaluru's buses carry.

CHECK · 40% OF WORKING PROFESSIONALS COMMUTE BY CAB DOESN'T HOLD UP

Bengaluru's city buses alone carried about 40 lakh passenger trips a day in 2025, and two-wheelers are the most common way to get to work. App cabs are a small share of daily trips, a few per cent. A 40% cab share would put more cars on the Outer Ring Road than it could physically hold.

STRONGER ANSWER



Figure 4.23 · The six cards, applied to a driver fleet. The trip cycle (card 4) and the per-driver check (card 5) are where fleet guesstimates go wrong. *Author's summary.*

A 1 · UNIT

I'll count distinct Uber car drivers who drive in Bengaluru on a typical weekday. Autos and bikes I'll add at the end, since Uber runs those too.

Why this move: "drivers" could mean registered, monthly active or online now; these differ by 3–5×, so pick one out loud.

A 2 · APPROACH

Supply from demand: daily trips, then the busiest hour, then how many trips one driver completes in that hour, then how many of the day's drivers are online at that hour.

Why this move: drivers are capacity, and capacity is set by the peak.

A 3–4 · TREE AND NUMBERS

1.4 crore people. About 2% take an app cab on a weekday (commuters, airport runs, late nights), about 1.8 trips each: $1.4 \text{ crore} \times 2\% \times 1.8 \approx 5 \text{ lakh cab trips}$. Uber's share of cabs is about 50%: 2.5 lakh. The busiest hour carries about 10% of the day: 25,000 trips. A trip cycle at peak, 35 minutes riding plus 10 to reach the pickup, is 45 minutes: 1.33 trips an hour. $25,000 \div 1.33 \approx 19,000$ drivers online at the peak. About 60% of the day's drivers are on then: $19,000 \div 0.6 \approx \mathbf{31,000 \text{ drivers}}$ (Figure 4.24).

Why this move: every number has a one-line reason, and the cycle includes the empty minutes that the candidate left out.

A 5 · CHECK

Per driver: $2.5 \text{ lakh trips} \div 31,000 \approx 8 \text{ trips a day}$, a little below a full-timer's 10–12, which fits a mix of full- and part-timers. Per resident: 5 lakh cab trips for 1.4 crore people is one app-cab trip per person every four weeks. Both sound right for Bengaluru.

Why this move: two quick ratios, one per side, catch almost any error in a fleet estimate.

A 6 · RANGE

The swing factor is the share riding each day: at 1%, about 16,000 drivers; at 3%, about 47,000. Monthly active drivers are higher, perhaps 1.5×, because few drive every day. With Uber Auto and Moto added, the whole fleet is roughly double.

Why this move: a range with its driver tells the interviewer which assumption to argue with.

HOW THE ANSWER HANGS TOGETHER

The unit (distinct car drivers on a weekday) fixed what to count. Supply from peak demand gave the tree, the 45-minute cycle made the capacity realistic, and two ratios, trips per driver and trips per resident, confirmed the scale. The range names the one input that moves it.

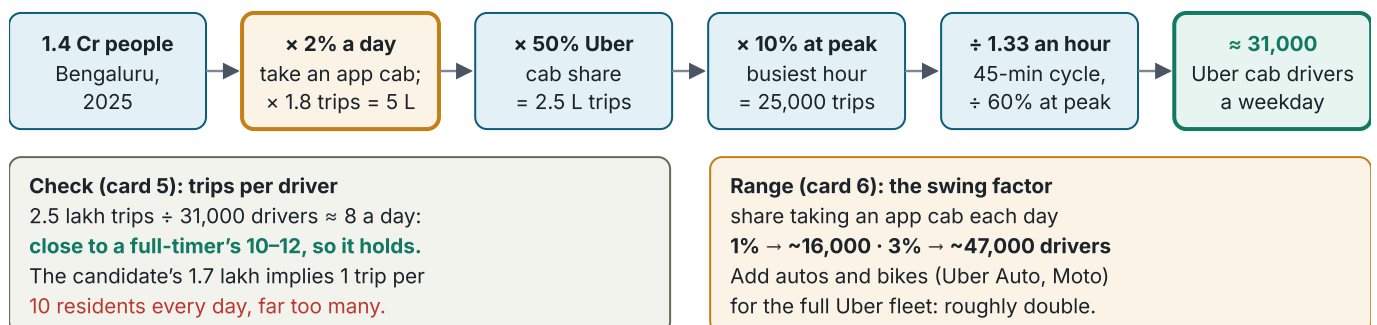


Figure 4.24 · Uber car drivers in Bengaluru, as an equation tree. The orange box is the swing factor; the green box is the answer; the grey box is the check that rejects the casebook's 1.7 lakh. *Population rounded from 2025 estimates; Uber cab share from industry estimates (2025); all other inputs illustrative.*

THE REUSABLE TRICK

Drivers needed at peak = peak trips an hour $\times$ cycle time in hours. $25,000 \text{ trips} \times 0.75 \text{ hours} = \text{about } 19,000$. This is Little's law: things in the system = arrival rate $\times$ time each one stays. Use it for drivers, riders, delivery bikes, checkout counters and servers.

GUESSTIMATE 4.2 · BIKES FOR A NEW BIKE-TAXI FIRM

Guesstimate

Medium

IIM B Sigma PM Casebook 2026–27, p. 140

Try it first. At what time of day would you most want a bike taxi, and how many other people want one at the same moment? Size the fleet for that moment, for five minutes.

THE QUESTION

You are starting a bike taxi company, how many bikes would you require? (Google PM)

- A** At first, we need to determine the scope of operations. Are we starting in a single city or planning to expand to multiple cities simultaneously, like pan-India?
- Q** We are just starting out in Bangalore for now.
- A** Okay, so the first step would be to forecast the demand in the city. We can start with the population of the city which is approx. 1.4 crore.
- Q** What other factors will you need to get to a number?
- A** We can consider other factors like population density, age and income segmentation of the population, existing public transport options, and even the competitor's market share in the city and how much market we can expect to capture.
- Q** How will you use these factors?
- A** We can start with segmenting by age. In India roughly 25% of the population falls in the 0-15 years age group, 40% in the 15-35 range, 25% in the 35-60 range, and the remaining 10% of the people are 60+ years old. Bike taxis will majorly be used by the 15-35 year old age group which includes college going students and working professional. So, the number of potential riders comes out to be $1.4\text{cr} \times 40\% = 56 \text{ lakh people}$. Then we can look at income segmentation. In a metro city like Bangalore the distribution of lower, middle, and upper-class people will be approx. 20%, 70%, and 10% respectively. I believe bike taxis will be a viable choice for the middle-class population which comes out to be $56 \text{ lakh} \times 70\% \sim 40 \text{ lakh people}$.
- Q** That seems awfully high. Will all 40 lakh people really use bike taxis for conveyance?
- A** No, we can further drill down by looking at the alternative transportation options that people have. Common ones being personal cars/bikes, cabs, autos, bike taxis, metro, and even bicycles. People generally prefer bike taxis for travelling over short distances (upto 10kms maximum) and considering that these bikes can carry only one passenger at a time, a person would prefer a bike taxi out of all available options maybe 10% of times whenever they want to travel. So, the number of rides comes out to be $40 \text{ lakh} \times 10\% = 4 \text{ lakh}$. Finally, we also need to consider the existing competition. There are many prominent companies already providing bike taxi services in Bangalore. Initially it will be difficult to capture a sizeable share of the market, but with the right kind of marketing and promotional offers, we can aim to capture 10% of the market. That comes out $4 \text{ lakh} \times 10\% = 40,000 \text{ rides daily}$.
- Q** How will you get to the number of bikes required from this?
- A** Considering that bike taxis will be used for distances upto 10kms, which might take 20-30 minutes on average to complete depending upon the traffic conditions, it is safe to say that a rider may only complete 2 rides in an hour. Also considering that a rider works 10 hours a day on average, the total number of trips completed by a rider in a day comes out to be 20. So, to get the total number of bikes required, we can simply divide $40,000 / 20$ to get 2000 bikes.
- Q** That seems alike a fair estimate. But what about maintenance, repairs, and having some bikes as backups?
- A** Yes, these are important considerations, and I would recommend adding around 10% to the estimated number of bikes to account for repairs and maintenance of some bikes. The total number of bikes required in that case will come out to be 2,200 bikes.
- Q** Seem practical. That would be all.

The casebook prints the candidate's three splits as tables at the end of the case, under the note "Segmentation breakup followed".

AgeGroup	Split	Income Group	Split	Transportation Mode	Preference
0-15	25%	Lower Class	20%	Walking	5%
15-35	40%	Middle Class	70%	Bicycle	5%
35-60	25%	Upper Class	10%	BikeTaxi	10%
60+	10%			Autos	20%
				Cab Services	20%
				Metro	15%
				Personal Car/Bike	25%

WHAT WORKED · WHAT WAS MISSING

- ✓ **Scoped to one city** before any number.
- ✓ **Segmented by age and income**, and cut the pool when challenged.
- ✓ **Converted demand into a fleet** with rides per bike per day, then added 10% for repairs.
- ✗ **Sized on the daily average.** Bike-taxi rides bunch at 8–10 am and 6–8 pm. A fleet built for 20 evenly spread rides a bike runs out at 9 am.
- ✗ **A unit jump.** “10% of the times they travel” became “10% of people, once a day”. Say the trips-per-person assumption out loud.
- ✗ **Never asked who owns the bikes.** Indian bike-taxi firms mostly sign up riders with their own bikes; the question may really be “how many riders?”.
- ✗ **No check** against a rider’s day or earnings.

Guesstimate kit check: 1 Unit ✗ **bikes owned or riders on the app?** · 2 Approach ✓ **demand to fleet** · 3 Tree ✓ · 4 Numbers with reasons ✓ · 5 Check ✗ · 6 Range ✗. Skipping card 1 left the ownership model open; skipping the peak halved the answer.

CHECK · “A RIDER WORKS 10 HOURS... 20 TRIPS IN A DAY” **OVERSTATED**

Twenty trips assumes the bike is busy every minute of ten hours. Real riders wait between rides, ride empty to pickups and take breaks. Driver surveys and platform statements usually suggest 10–15 trips on a full day. At 12 trips, the same 40,000 rides need about 3,300 riders, not 2,000.

STRONGER ANSWER

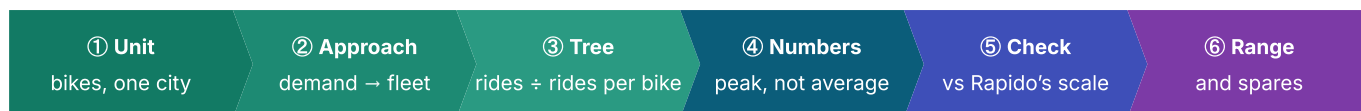


Figure 4.25 · The six cards, applied to a bike fleet. Card 4 uses the peak hour, not the daily average. *Author's summary.*

- A 1 · UNIT**
Bikes active on a weekday in Bengaluru. I'll assume riders bring their own bikes, as at Rapido, so the count equals riders we must sign up. If we own an electric fleet, I'll add spares for charging and repairs.
Why this move: the ownership model decides whether the answer is a hiring target or a purchase order.
- A 2 · APPROACH**
Demand side, to the busiest hour, then capacity: peak-hour rides ÷ rides one bike completes in an hour.
Why this move: a fleet that works at 9 am works all day; the reverse is not true (Figure 4.26).
- A 3–4 · TREE AND NUMBERS**
I'll keep the candidate's pool: 40 lakh middle-income 15–35-year-olds. If each makes one trip a day and 10% of those go by bike taxi: 4 lakh rides a day across all firms. A new firm wins 10%: 40,000. The morning peak hour carries about 15% of the day: $40,000 \times 15\% = 6,000$ rides. A trip of about 20 minutes plus the ride to the pickup is a 40-minute cycle: 1.5 rides an hour. $6,000 \div 1.5 = 4,000$ bikes, and 4,400 with 10% spare (Figure 4.27).
Why this move: the peak share and the cycle time are the two numbers the candidate skipped; together they double the fleet.
- A 5 · CHECK**
From the rider's side: a full day is realistically about 12 rides. $40,000 \div 12 \approx 3,300$ riders, the same range. And a rider doing 12 rides at about ₹60 earns about ₹720 a day before fuel, a believable gig income, so signing up 3,000–4,000 riders in a city of 1.4 crore is plausible.
Why this move: the rider's day is a second route to the same number, and the earnings check tests whether that supply would actually show up.

A 6 · RANGE

The swing factor is how peaky demand is. If the peak hour carries 10% of rides, about 2,700 bikes; at 20%, about 5,300. So 3,000–5,000 bikes, and I'd plan to sign up 4,000 riders, with incentives for the 8–10 am slot.

Why this move: naming the peak share as the swing factor turns the estimate into an operating plan: pay extra for the hour that sets the fleet size.

HOW THE ANSWER HANGS TOGETHER

Card 1 turned "bikes" into "riders with their own bikes". The tree kept the candidate's demand but sized the fleet for the 9 am peak, with a realistic trip cycle. The rider's-day check lands in the same range, and the range points to the peak share, which is also where the firm should spend its incentives.

Share of a day's bike-taxi rides in each hour, Bengaluru weekday (illustrative)

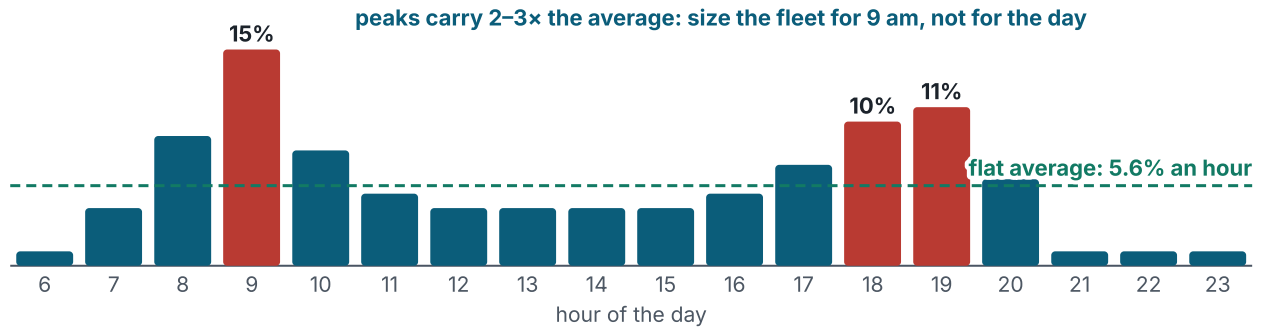


Figure 4.26 · Why the average misleads. The busiest morning hour carries almost three times the average hour's rides. A fleet sized on the average would leave most 9 am riders waiting. *Illustrative weekday profile, shaped like published ride-hailing demand curves for Indian metros.*

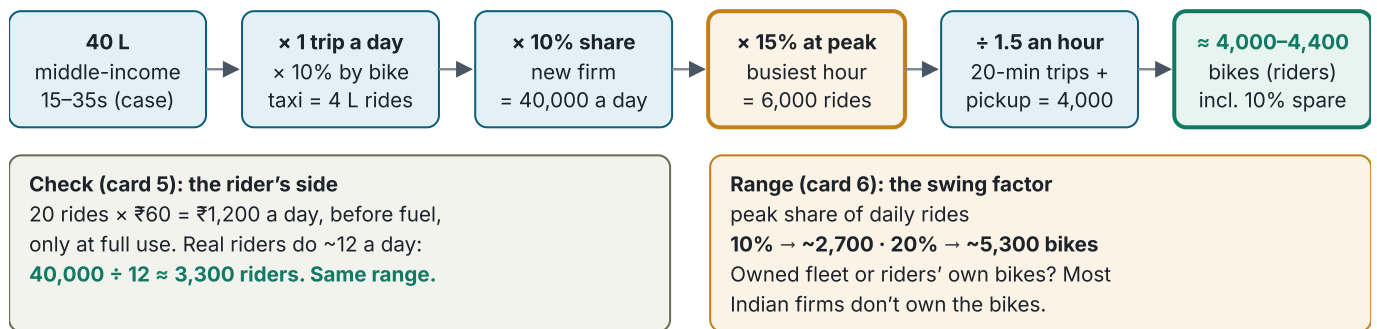


Figure 4.27 · Bikes for a new bike-taxi firm, as an equation tree. The peak share (orange) moves the answer most; the rider's-day check lands in the same range. *Demand pool from the casebook; peak share, cycle time and fares illustrative.*

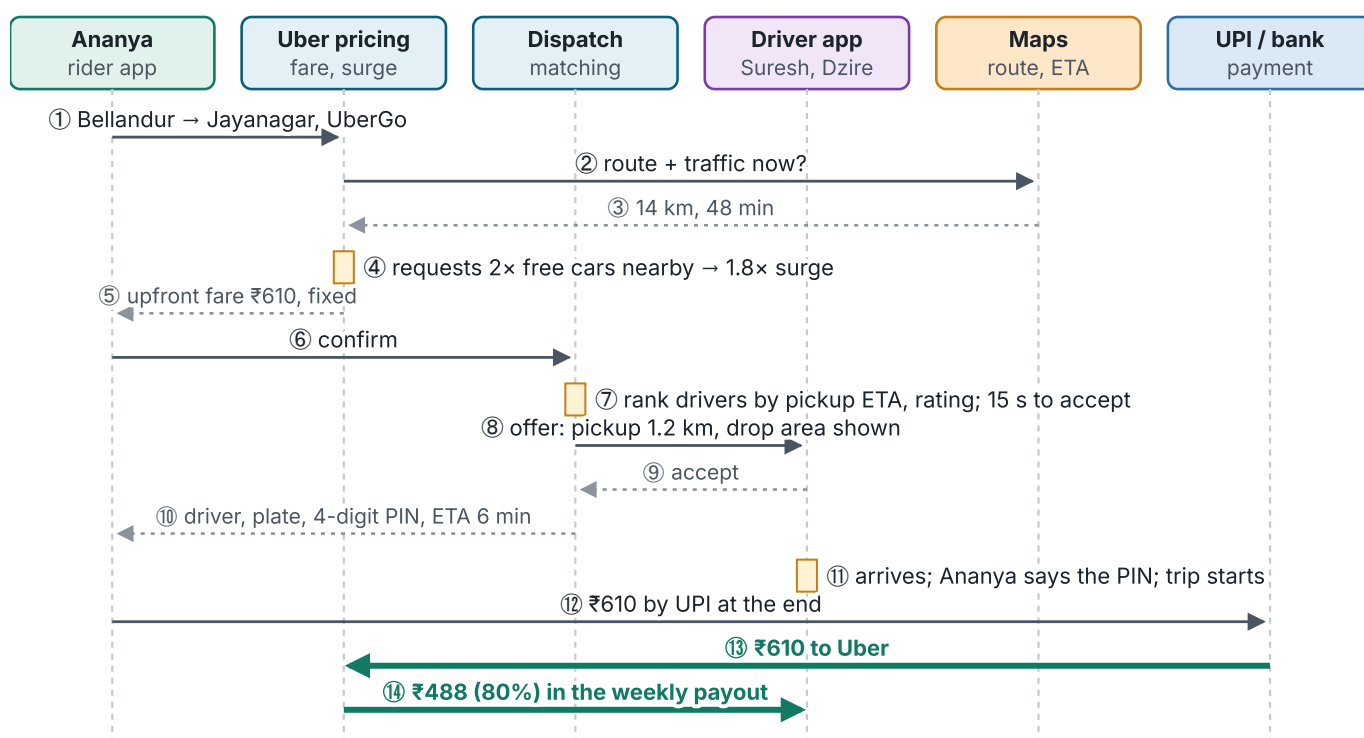
THE REUSABLE TRICK

Peak-hour share ≈ 10–15% of the day for commuting products (rides, metro, office lunch delivery). Fleet ≈ daily trips × peak share × cycle time in hours. $40,000 \times 15\% \times \frac{2}{3} \text{ hour} = 4,000$. One line, and it works for buses, delivery riders and call-centre agents.

Uber: a pricing and matching engine that happens to move cars

Walkthrough (illustrative persona). Ananya, from the cold open, finally gets a car. She types Jayanagar; the app shows an upfront fare of ₹610 at 1.8× surge before she commits. She confirms. In the next two seconds Uber's dispatch ranks nearby drivers by how soon each could reach her, and offers the ride to Suresh, 1.2 km away, who sees the drop area and accepts. Ananya gets his name, plate and a four-digit PIN. When he arrives she reads out the PIN, the trip starts, and at the end ₹610 leaves her UPI account. Suresh sees ₹488 in his weekly payout (Figure 4.28).

Feature	User problem	Why it's designed this way	Metric it moves
Upfront fare	"Will the meter cheat me in traffic?"	A fixed price before booking; Uber, not the rider, carries the risk of a jam	Booking conversion; fare disputes
Ride PIN	"Is this the right car?"	The trip starts only when the rider's PIN is entered; built for India first	Wrong pickups; safety reports
Auto, Moto, Go, Premier, Intercity	"A cab is too expensive for 3 km"	A price point for every trip; autos moved to a driver subscription in 2025	Trips per rider; share of short trips
Share trip, SOS	"My family worries at night"	Live location to contacts; a safety line inside the app	Night-time trips; women riders
Uber One, Uber for Business	"I ride often"; companies need billing	Membership perks; central invoices for employees	Frequency; corporate trips



Surge doubles as a message to drivers: "come to Bellandur". The PIN stops wrong-car pickups.

Figure 4.28 · Behind Ananya's ride. Pricing (④) and matching (⑦) happen before any car moves. The surge fare earns Uber more, but its main job is to pull drivers into Bellandur. *Simplified from Uber's published rider and driver flows; amounts illustrative; 80% driver share per India's 2025 aggregator guidelines.*

Core loop: liquidity

Uber's whole strategy is one loop (Figure 4.29). More riders in an area mean drivers earn more per hour, so more drivers log on. More drivers mean a car is always close, so pickups are quicker and drivers spend less time driving empty. Less empty driving means Uber can hold fares steady and still pay drivers well, which brings more riders. The loop is **local**: winning Mumbai does nothing for Indore. That is why rivals can attack city by city.

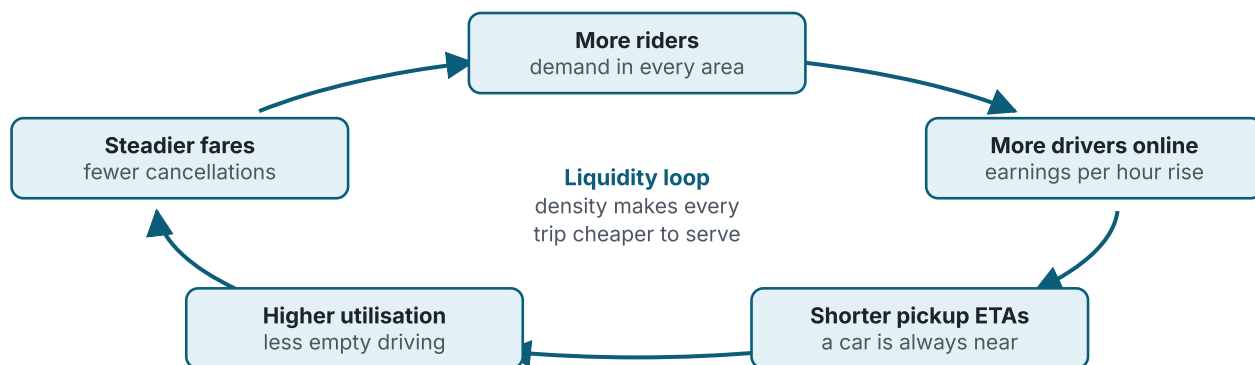


Figure 4.29 · The liquidity loop. Density is the product. A city where Uber has twice the drivers per square km has shorter ETAs and lower costs per trip. *Author's diagram, after Uber's published growth logic.*

Business model, with numbers

Uber doesn't report India separately. Worldwide, in April–June 2026, riders and eaters paid **\$58.0 bn** (+22%); Uber kept **\$14.2 bn** as revenue and earned **\$2.8 bn** of adjusted EBITDA, on 3.9 bn trips and 208 mn monthly users (Figure 4.30). In India it runs two models side by side: commission on cabs (capped at 20% for driver-owned cars under the 2025 rules) and a daily subscription for autos since February 2025.

Uber worldwide, April–June 2026



Figure 4.30 · From bookings to profit, one quarter. Of every \$100 riders pay, Uber keeps about \$24 as revenue and about \$5 as adjusted EBITDA; the rest goes to drivers, couriers and restaurants. *Uber Q2 2026 results (August 2026).*

Moat, in plain words. Density in each big city, which makes pickups faster than a smaller rival can offer. Global pricing, mapping and safety technology that a local player can't afford to build. And a brand that corporate travel desks trust. The weakness: drivers and riders both use several apps, so the moat must be re-earned every evening.

NORTH STAR Completed trips per week.	INPUTS Pickup ETA · request-to-completion rate · drivers online at peak · utilisation.	GUARDRAILS Driver and rider cancellations · safety incidents per million trips · driver earnings per hour · surge share.
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2025–26 moves. Subscription for auto drivers (February 2025). Compliance with the July 2025 guidelines: 80% to drivers on own vehicles, surge up to 2× base. Bike taxis restarting in Karnataka after the January 2026 court order. Metro tickets and bus shuttles inside the app in some cities, to own more of the commute.

STRENGTHS

- ✓ Market leader in cabs and autos
- ✓ Profitable worldwide, so it can outspend locals
- ✓ Trusted by companies and airport travellers

WEAKNESSES

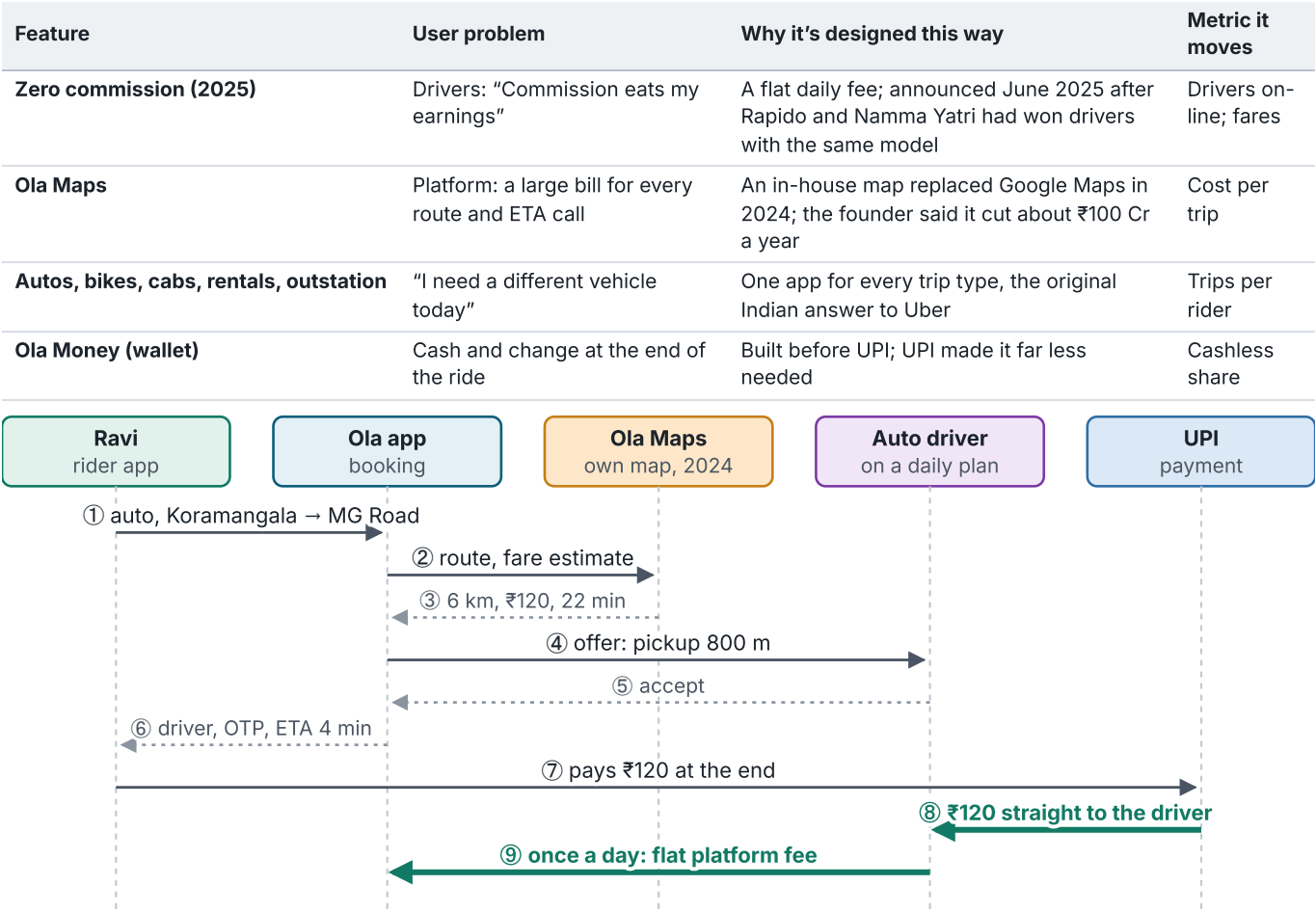
- ✗ Priced above Rapido and Namma Yatri for autos and bikes
- ✗ Cancellations and surge anger riders at peaks
- ✗ Rules on fares and fees now vary state by state

What I'd build next

- Office-goers pay surge on the same route at 9 am every day → a weekly commute pass: fixed route and time, a fixed price below average surge, guaranteed pickup → weekday trips per pass holder; driver utilisation at peak.
- Drivers cancel trips that end far away → a return premium on edge trips, offered first to drivers heading that way → completed trips per request on edge trips.
- Families don't trust bike taxis for teenagers → a "trusted rider" mode: verified drivers with top ratings, live sharing to a parent, no cash → share of under-21 riders; safety reports.

Ola: how a market leader loses a two-sided market

Walkthrough (illustrative persona). Ravi, 34, books an Ola auto from Koramangala to MG Road. Ola’s own map, Ola Maps, estimates 6 km, ₹120 and 22 minutes. A driver 800 m away accepts; Ravi gets an OTP and an ETA of 4 minutes. At the end he pays ₹120 by UPI, and all of it goes to the driver. Ola earns nothing on this ride. It earns once a day, from the flat fee the driver pays to stay on the platform (Figure 4.31).



Under zero commission Ola earns per driver-day, not per ride; its own map removed the per-call maps bill.

Figure 4.31 • Behind Ravi’s auto ride. The money (⑧) now skips Ola entirely; Ola’s income (⑨) depends on how many drivers pay the daily fee, not on how many rides happen. *Simplified; fare illustrative; zero-commission model announced June 2025.*

The loop, running backwards

Ola had more of India’s cab market than Uber in the mid-2010s. By FY25 its ride-hailing revenue had fallen 42% to ₹1,171 Cr, with a loss of ₹662 Cr; a write-down of its Ola Electric stake pushed the group loss to about ₹1,975 Cr. The causes are a lesson in two-sided markets. Leadership time and money went to Ola Electric and the AI venture Krutrim. Drivers complained about payouts and incentives and drifted to rivals. Fewer drivers meant longer waits and more cancellations, and riders followed. The same loop that grows a market-place shrinks it just as fast (Figure 4.32).

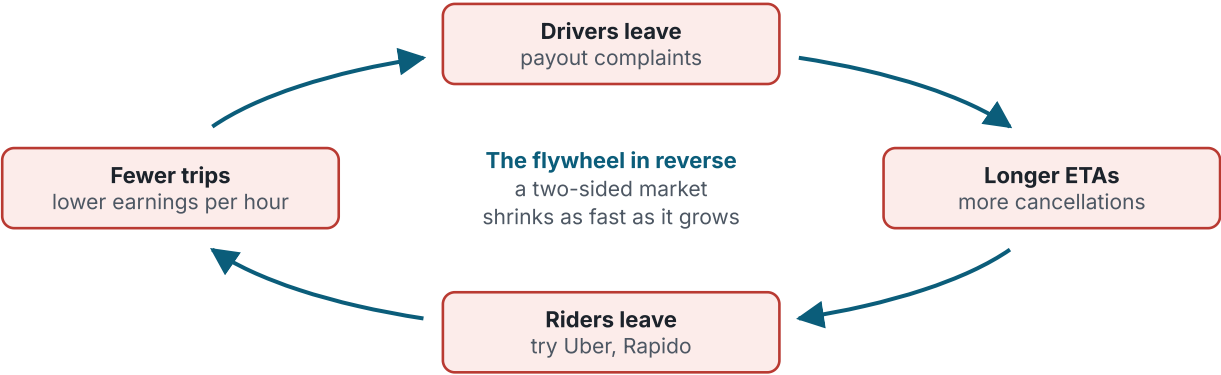


Figure 4.32 • The flywheel in reverse. Lose drivers and riders wait longer; lose riders and drivers earn less. Each turn makes the next one easier. *Author’s diagram; revenue and loss from Ola Consumer’s FY25 filings, reported May 2026.*

Business model. Now mainly flat fees from drivers, plus fees on cab rentals and outstation trips. Revenue no longer rises with rides, so it must rise with active drivers or new products. **Moat, in plain words.** Thin today: a known brand and its own map. The map is a real asset: it removes a cost every rival pays and could be sold to other companies.

NORTH STAR Completed trips per week in its top five cities.	INPUTS Paying drivers online at peak · pickup ETA · repeat riders.	GUARDRAILS Driver payout complaints · cancellations · cash burn per trip.
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2025–26 moves. Zero commission (June 2025); Ola Maps offered to outside developers; board approval to prepare an IPO (2026 filings), with a loan of about \$65 mn due in December 2026.

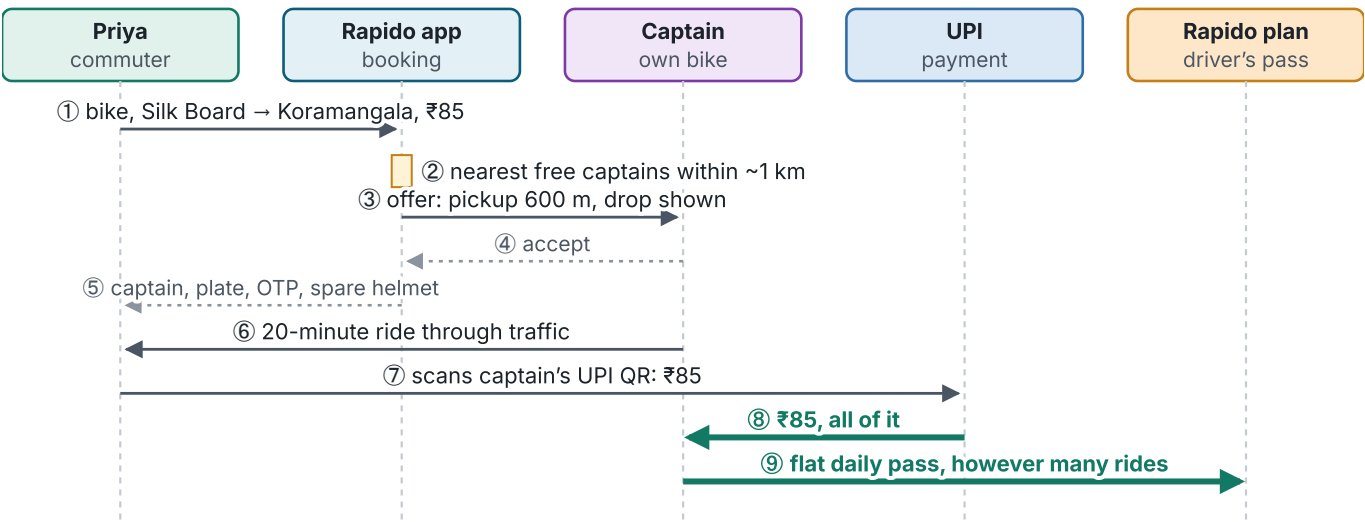
STRENGTHS ✓ Its own map: a lower cost per trip ✓ A brand every Indian rider knows ✓ All vehicle types in one app	WEAKNESSES ✗ Lost driver trust; weak service reputation ✗ Late to subscriptions, so no counter-position left ✗ Thin cash for a price or incentive war
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What I'd build next (a turnaround)

- 1. Supply is the constraint → a guaranteed minimum hourly earning for 90 days in five core cities, funded by pausing weak cities → drivers online at peak; ETA.
- 2. Returning riders remember bad experiences → an automatic ₹50 credit whenever a driver cancels → 30-day retention of returning riders; driver cancellations.
- 3. Ola Maps is a cost saver nobody outside uses → sell it to delivery and logistics firms with Indian-address features → API revenue; external developers.

Rapido: the bike taxi that changed how drivers are paid

Walkthrough (illustrative persona). Priya, 24, is stuck at Silk Board at 8:40 am with a 9:30 meeting. A cab would take 50 minutes and cost ₹400. She books a Rapido bike for ₹85. A “captain” (Rapido’s word for its bike riders) 600 m away accepts, hands her a spare helmet, checks her OTP and weaves through traffic. Twenty minutes later she scans the captain’s UPI QR code. All ₹85 goes to the captain, who pays Rapido a flat daily pass instead of a commission (Figure 4.33).



A cab would take 50 minutes and ₹400. Time saved, not the fare, is why bike taxis win at Silk Board.

Figure 4.33 · Behind Priya's bike ride. The fare goes straight to the captain (⑧); Rapido earns from the pass (⑨). The spare helmet (⑤) is a safety and a legal requirement. *Simplified; fare illustrative; subscription model as described by Rapido and press, 2025–26.*

Feature	User problem	Why it's designed this way	Metric it moves
Bike taxi	"Traffic makes cabs slow and expensive"	A two-wheeler slips through jams at a fraction of a cab's fare	Trips per rider; time saved
Driver subscription	Captains: "I lose a quarter of every fare"	A flat daily pass; captains keep the whole fare, so fares can be 10–15% lower	Captains online; fares
Autos and cabs	"Sometimes I need a bigger vehicle"	Cross-sell to riders who already trust the app	Share of non-bike trips

Parcels and Ownly (food)	Captains idle between peaks; restaurants hate high commissions	The same fleet carries parcels and food in quiet hours	Off-peak utilisation
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Core loop and the product ladder

The loop starts with the cheapest, fastest ride in a traffic jam. A commuter who saves 30 minutes every morning comes back daily, which no cab can match for frequency. Captains who pay no commission earn more per ride and stay loyal. From that base Rapido climbs a ladder of products that reuse the same app, riders and captains (Figure 4.34).

Rapido's product ladder: each step reuses the same app, riders and captains
Each rung raises trips per rider, or keeps captains busy in quiet hours.

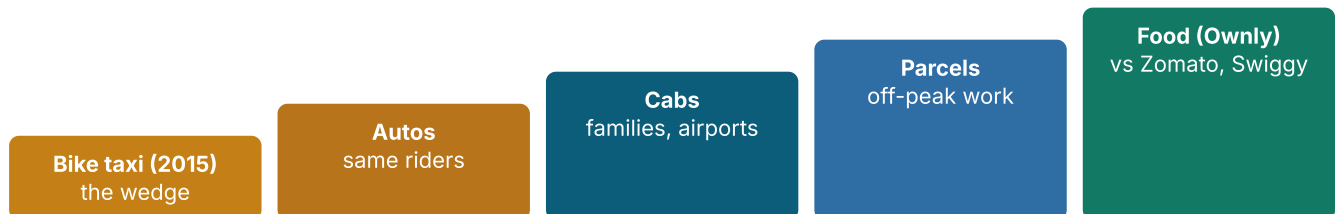


Figure 4.34 · One fleet, five products. Bikes win the daily habit; autos and cabs raise spend per rider; parcels and food fill the captains' quiet hours. *Company announcements, 2015–2026.*

Business model, with numbers. FY25 revenue from operations was **₹934 Cr** (+44%) and the loss narrowed to **₹258 Cr** from ₹370 Cr. In May 2026 it raised \$240 mn at a **\$3 bn** valuation, led by Prosus; its co-founder has said it was profitable for one quarter in FY25 and is preparing for an IPO. The subscription is **counter-positioning**: Uber and Ola could match it only by giving up commission, and for autos they did.

REAL-LIFE EXAMPLE · NAMMA YATRI, THE DRIVER-OWNED COMPARISON

Namma Yatri started in Bengaluru in 2022 on the open ONDC and Beckn rails, with auto drivers' unions involved from the start. Drivers pay ₹25 a day for unlimited rides (or ₹3.50 a ride for the first ten) and keep every rupee of the fare. By August 2026 it had 7.7 lakh drivers enabled. It proved that a thin, open platform could win drivers, and within two years every major app had moved autos to a fee model. Its limit: with so little revenue per ride, it has little to spend on rider marketing or safety operations.

Moat, in plain words. Captain loyalty (no commission), the daily commuting habit, and the lowest fares. The risk is regulation: Karnataka stopped bike taxis in June 2025; the High Court lifted the ban in January 2026 and the state has appealed to the Supreme Court.

NORTH STAR Completed trips per week.	INPUTS Paying captains online at peak · pickup ETA · share of trips outside bikes.	GUARDRAILS Safety incidents · regulatory status by state · captain churn.
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2025–26 moves. Ownly food delivery with low restaurant commissions (2025); the \$3 bn round (May 2026); bike taxis back in Bengaluru after the January 2026 court order.

STRENGTHS

- ✓ Leader in bike taxis, about 56% of rides
- ✓ Loyal captains; the lowest fares
- ✓ A daily habit, not an occasional trip

WEAKNESSES

- ✗ Bike taxis depend on each state's rules
- ✗ Little revenue per ride; must earn from volume
- ✗ Food puts it against Zomato and Swiggy

What I'd build next

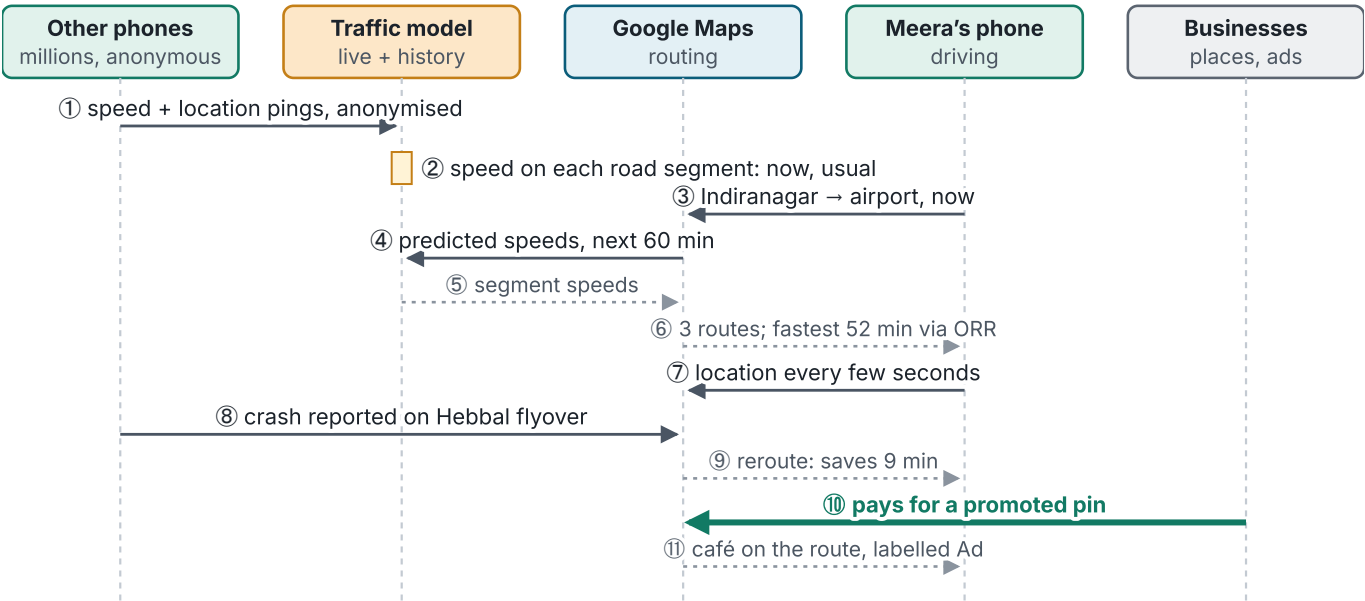
1. Many women avoid bike taxis → an opt-in women-captain option with live trip sharing → share of women riders; safety reports.
2. Captains want to own electric bikes but lack credit → partner loans repaid through the daily pass → captains financed; 12-month captain retention.
3. Regulation can switch off a city overnight → a state-by-state readiness plan: shift captains to autos and parcels within a week of a ban → captain earnings kept during a ban.

Google Maps: two billion users who are also its sensors

Walkthrough (illustrative persona). Meera, 41, needs to reach Bengaluru airport from Indiranagar for a 9 pm flight. At 6:30 pm Maps offers three routes; the fastest takes 52 minutes via Outer Ring Road. Twenty

minutes in, someone reports a crash on the Hebbal flyover; Maps reroutes her and saves nine minutes. A café on the new route appears as a labelled ad. None of this needs a single traffic camera: the speeds come from millions of other phones moving through the same roads (Figure 4.35).

Feature	User problem	Why it's designed this way	Metric it moves
Live traffic and ETA	"How long will it really take?"	Anonymous speed pings from phones, mixed with each road's usual speed at that hour	ETA accuracy; routes started hour
Address descriptions (India, 2024)	"The pin is wrong; which gate?"	Landmarks near the address, because many Indian addresses are descriptive	Arrivals at the right door
Ask Maps (Gemini, March 2026)	"Where can I eat near the metro, not crowded, under ₹500?"	Plain-language questions answered from places, reviews and photos	Discovery searches; saves
Metro tickets via ONDC	"One more app just for a metro ticket"	Buy tickets inside Maps in some cities, through India's open commerce network	Transit trips planned in Maps
Reviews, photos, Local Guides	"Is this place any good?"	Users add the content; points and badges reward them	Place coverage; review freshness



Every driver is both a user and a sensor: the ETA gets better as more people use it.

Figure 4.35 · Behind Meera's drive. Other users' phones (①, ⑧) produce the traffic picture; Meera's phone (⑦) feeds it back. Businesses (⑩) pay to be seen along the route. *Simplified from Google's published descriptions of traffic prediction; timings illustrative.*

Core loop: data

Maps improves because people use it (Figure 4.36). More drivers mean more speed readings on more roads, so ETAs get more accurate, so more people trust Maps and use it. The same holds for places: more users write more reviews and upload more photos, which make search results better. A new map can copy the roads; it can't copy a decade of live readings and reviews.

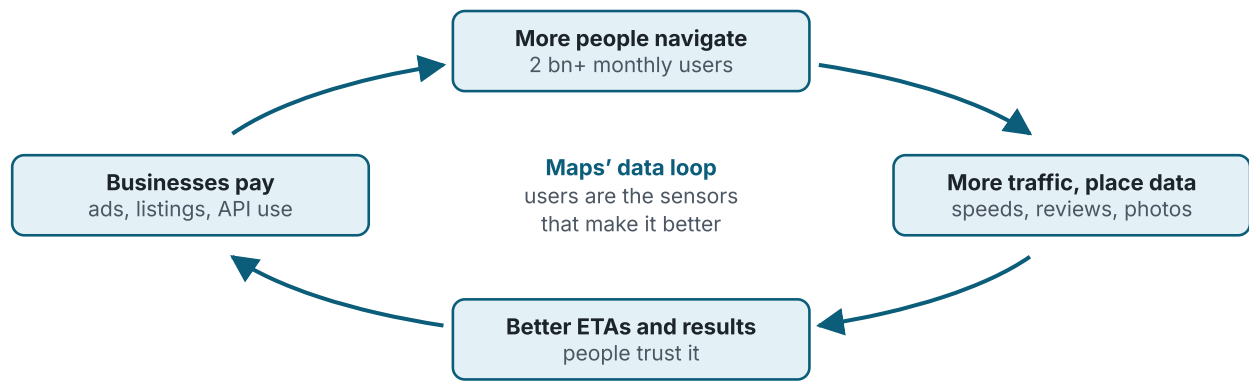


Figure 4.36 · The data loop. Every user is also a sensor and a reviewer, so the product gets better for free as it grows. *Author's diagram; 2 bn+ monthly users per Google (2025–26).*

Business model. Maps is free to users. Alphabet doesn't report its revenue, but it earns in three ways: **local ads** (promoted pins and sponsored results when people search “near me”); **Google Maps Platform** (companies pay per 1,000 calls to show maps, routes and places in their own apps); and **feeding Search**, where local intent makes ads more valuable. The biggest customers of the Platform, ride and delivery apps, are also the ones most likely to build their own map, as Ola did.

Moat, in plain words. The live data from two billion users, years of Street View imagery and place reviews, and the default spot on most Android phones.

NORTH STAR Trips and place decisions completed with Maps each week.	INPUTS ETA accuracy · share of searches ending in a visit or direction · fresh reviews per place.	GUARDRAILS Wrong-place reports · ad share of local results · privacy complaints.
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2025–26 moves. Ask Maps and Immersive Navigation (a 3D, lane-level view) launched in the US and India in March 2026; Ask Maps reached 150+ more countries in August 2026. India-specific features: narrow-road alerts, flyover callouts and address descriptions (2024).

STRENGTHS

- ✓ Unmatched live traffic and place data
- ✓ Default on Android; 2 bn+ users
- ✓ AI answers built on real reviews

WEAKNESSES

- ✗ Indian addresses and pins still go wrong
- ✗ Big API customers build their own maps
- ✗ Ads in local results can erode trust

What I'd build next

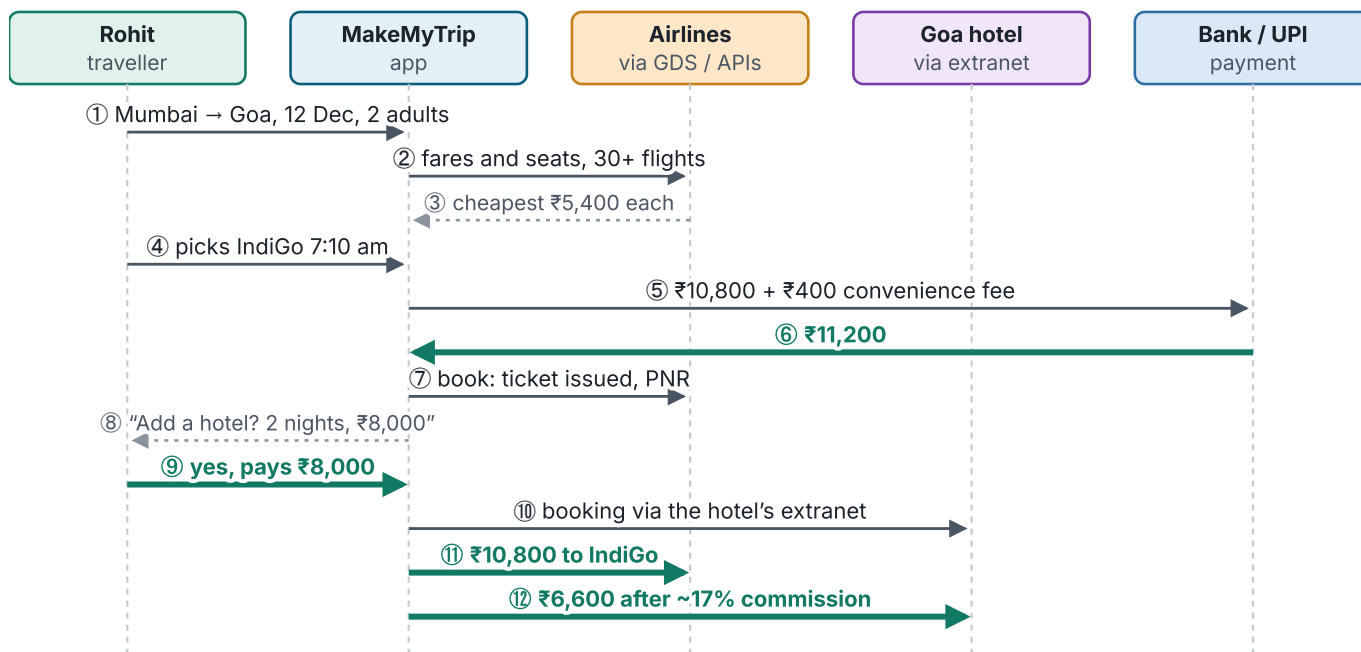
1. Delivery riders lose minutes finding the right gate → gate-level pins learned from where past deliveries actually ended → arrival-to-door time for Platform customers.
2. Tourists don't know what's worth seeing in two days → “Plan my day” from saved places, open hours and travel time → trips with a plan followed.
3. Commuters switch between metro, auto and walking → one ticket and one route across modes via ONDC → multi-mode trips booked in Maps.

MakeMyTrip: the flight brings you in, the hotel pays the bills

Walkthrough (illustrative persona). Rohit, 30, plans a December weekend in Goa with his wife. MakeMyTrip compares over 30 flights in seconds; he picks an IndiGo flight at ₹5,400 each and pays ₹11,200 including a convenience fee. The ticket is issued and a PNR appears. The next screen asks: “Add a hotel?” He books two nights at ₹8,000. The flight earned MakeMyTrip about ₹400; the hotel, about ₹1,400 (Figure 4.37).

Feature	User problem	Why it's designed this way	Metric it moves
Flight search and fare alerts	“Prices change every hour”	Fares from airlines and global distribution systems (GDS: the networks airlines use to sell through agents) in one list	Search-to-book conversion
Hotel and homestay cross-sell	“Where do we stay?”	Asked right after the flight, when intent is highest; hotels pay far more commission	Hotels per flight booking

Holiday packages	"Plan the whole trip for me"	Flights, hotels and transfers bundled; margin on the bundle	Revenue per customer
redBus, trains, cabs	"We're going by bus this time"	Covers the cheaper, more frequent trips	Trips per customer a year
myBiz (corporate travel)	Companies need policy and invoices	Approval rules and GST invoices built in	Corporate bookings



The flight earned MMT about ₹400; the hotel about ₹1,400. The flight brought Rohit in.

Figure 4.37 · Behind Rohit's Goa booking. Most of the flight money passes straight through to IndiGo (⑪); the hotel commission (⑫) is where MakeMyTrip earns. *Simplified; fares, fee and commission rate illustrative, typical of Indian OTAs, 2025–26.*

Core loop and business model

Flights are the most searched travel product and the least profitable. They bring people to the app often; each visit is a chance to sell a hotel, a package or a bus ticket, which pay more. More customers attract more hotels, which want to be listed where the travellers are, which gives customers more choice. In FY26 (to March 2026), MakeMyTrip handled **\$10.4 bn** of bookings (+10.4% in constant currency), passed **\$1 bn** of revenue for the first time and earned an operating profit of **\$156 mn**. In April–June 2026, bookings grew 19.9% to \$2.85 bn, though international travel was subdued by the West Asia conflict. The fastest-growing lines are the non-flight ones (Figure 4.38).

Growth in adjusted margin by segment, FY26 (to March 2026), constant currency

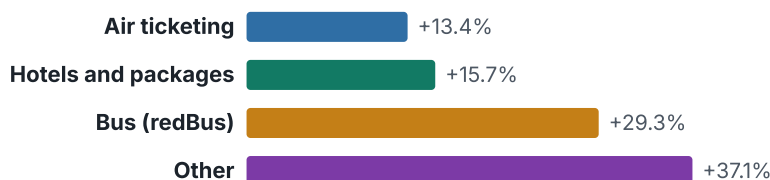


Figure 4.38 · Where profit growth came from in FY26. Buses (redBus) and other services grew fastest; hotels and packages, the biggest profit line, grew faster than flights. *MakeMyTrip FY26 results (May 2026), adjusted margin growth in constant currency.*

Moat, in plain words. Relationships with tens of thousands of Indian hotels and homestays; redBus, the leader in online bus tickets; a brand that most Indian travellers try first; and corporate accounts through myBiz. Airlines can sell direct, hotels much less so.

NORTH STAR Trips booked by repeat customers.	INPUTS Search-to-book conversion · hotels per flight booking · share of bookings with a package.	GUARDRAILS Refund and cancellation complaints · margin per booking · hotel payment delays.
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2025–26 moves. An experiences-booking platform (tours and activities, 2025); more AI trip planning in the app; growth in the UAE business for residents, including Umrah packages.

STRENGTHS

WEAKNESSES

- ✓ Largest Indian hotel supply
- ✓ redBus and myBiz widen the base
- ✓ Profitable and growing

- ✗ Airlines push travellers to book direct
- ✗ Google and AI assistants can intercept the search
- ✗ International travel is exposed to shocks

What I'd build next

1. Flyers book the hotel elsewhere → a “trip pack” price that drops when flight and hotel are booked together within 24 hours → hotels per flight booking.
2. Refunds after cancellations take weeks → instant refund to UPI for low-risk bookings, recovered from the airline later → repeat bookings within 90 days; complaints.
3. Indians in the Gulf fly home often → a “going home” hub: baggage-clear fares, festival alerts, family packages → repeat UAE customers.

Side by side

	Uber	Ola	Rapido · Namma Yatri	Google Maps	MakeMyTrip
For whom	Riders who value reliability; companies	Price-aware riders in every city tier	Commuters in traffic; auto riders	Everyone who moves or searches locally	Leisure and business travellers
Earns from	Commission (cabs), subscription (autos), Uber One	Driver daily fees	Driver passes; parcels, food	Local ads, Maps Platform API	Hotel commission, packages, fees
Latest number	Worldwide GB \$58.0 bn a quarter (Q2 2026)	FY25 revenue ₹1,171 Cr (-42%)	Rapido FY25 ₹934 Cr; NY 7.7 lakh drivers	2 bn+ monthly users	FY26 bookings \$10.4 bn
Moat	Density in big cities; global tech	Own map; brand	Driver loyalty; lowest fares	Live data from users	Hotel supply; redBus
North Star	Completed trips	Completed trips in core cities	Completed trips	Trips and decisions made with Maps	Repeat-customer trips
Biggest threat	Cheaper subscription apps	Running out of drivers and cash	State bans on bike taxis	AI assistants; own-map rivals	Direct airline sales; AI search

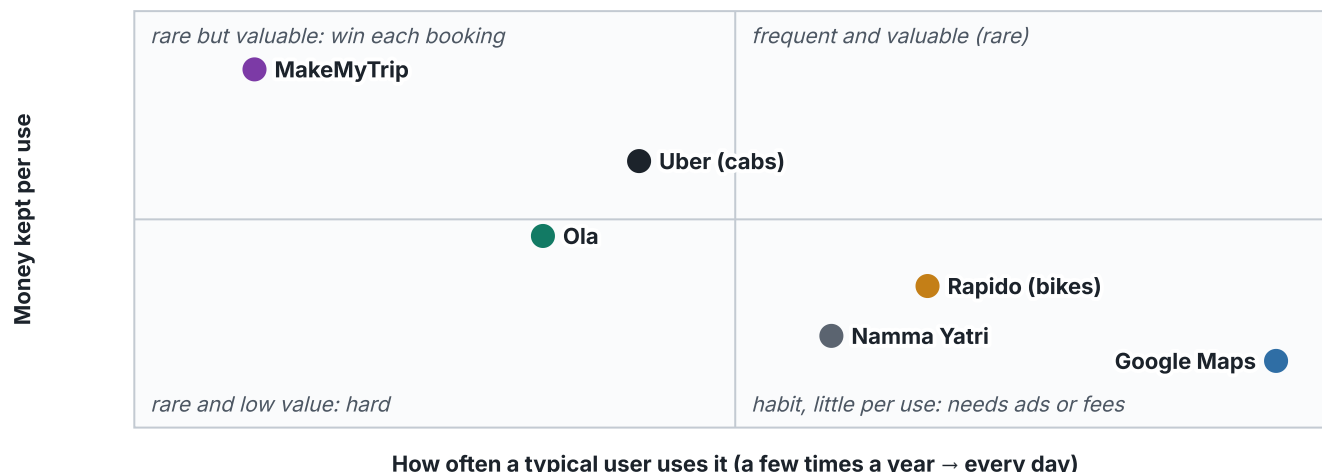


Figure 4.39 · How often people use it, and what the platform keeps each time. Travel is rare but valuable, so MakeMyTrip must win each booking. Maps is daily but earns almost nothing per use, so it needs ads. Ride apps sit in between, and subscriptions have pushed Rapido and Namma Yatri down: more habit, less money per ride. *Author's placement; positions are judgement, not measured values.*

IN THE INTERVIEW · HOW THESE PRODUCTS COME UP

- **“Why did Ola lose to Uber and Rapido?”** Drivers first: payout trust fell, drivers left, waits grew, riders followed. A two-sided market shrinks as fast as it grows.
- **“Is Rapido’s subscription model sustainable?”** It wins drivers and cheap fares, but revenue doesn’t grow with rides, so it needs volume products: autos, cabs, parcels, food.
- **“How does Google Maps make money?”** Local ads when people search nearby, fees from apps that use its maps, and more valuable ads in Search.
- **“Why do travel apps push hotels?”** A flight earns a few hundred rupees; a hotel pays 15–20%.

Cloud computing: renting the computer instead of owning it

On an ordinary Tuesday a ride app in India might need 40 servers. At midnight on New Year's Eve it needs 130, for about two hours. In 2005 a company would have bought 130 servers, paid for power and cooling all year, and watched 90 of them sit idle on 364 days. Today it rents them from Amazon Web Services (AWS), Microsoft Azure or Google Cloud by the second, and hands them back at 2 am. A PM doesn't configure any of this, but decides what must stay up, how much the bill may grow and which provider the product depends on.

The idea: computing as a utility

A century ago, big factories ran their own power plants. When the grid arrived, they plugged in and paid per unit. **Cloud computing** is the same move for computers: servers, storage, databases and AI models delivered over the internet, paid for as you use them. Three things change for a product team. You pay as you go instead of buying up front (running cost, not capital cost). You can add capacity in minutes, called **elasticity**. And someone else replaces the failed disk at 2 am. The providers are few and large: in April–June 2026, AWS had about 28% of the world market, Azure 20% and Google Cloud 15% (Synergy Research). All three run data centres in Mumbai and Hyderabad or Delhi, which matters for Indian rules that keep payment data in India.

How much do you want to run yourself? IaaS, PaaS, SaaS

The classic analogy is pizza (Figure 4.40). Make it at home and you do everything, from the oven to the dough. Take-and-bake gives you the kitchen and ingredients; you still cook. Delivery hands you a finished pizza. Dining out means you only eat. **IaaS** (infrastructure as a service) rents you virtual machines, storage and network; your team installs and runs everything above them. **PaaS** (platform as a service) also runs the operating system and scaling, so you bring only your code and data. **SaaS** (software as a service) is finished software used through a browser: Gmail, Zoho, Freshdesk.

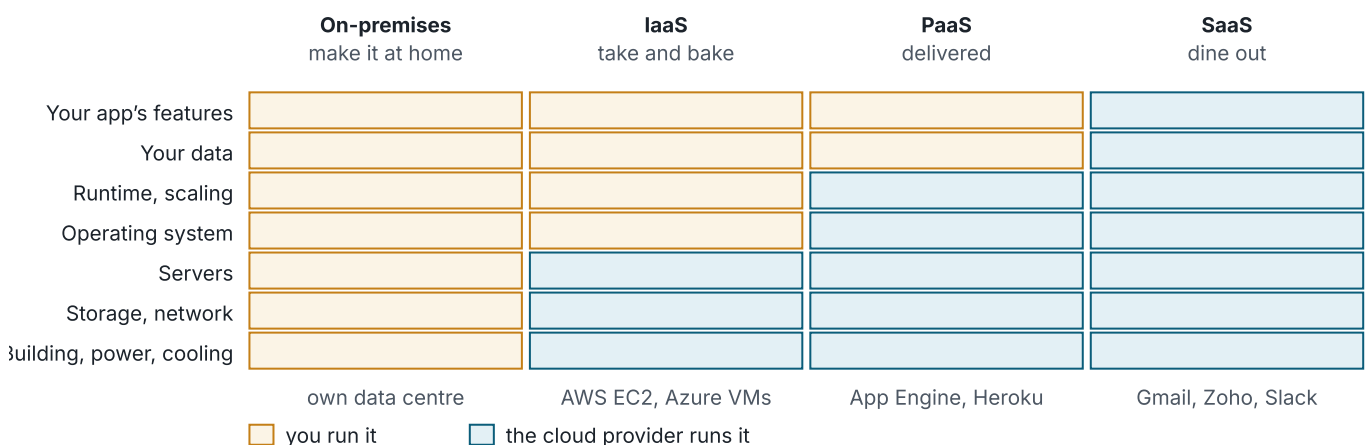


Figure 4.40 · Who runs which layer. Moving right gives you speed and less work; moving left gives you control and more work. A ride app runs its matching engine on IaaS or containers, uses PaaS databases, and buys SaaS for email and support tickets. *Author's diagram after the common "pizza as a service" analogy; examples current in 2026.*

Two newer rungs. **Containers** (Docker) package an app with everything it needs, so it runs the same on a laptop and in the cloud; **Kubernetes** runs thousands of containers and restarts or adds them automatically. **Serverless** (AWS Lambda, Google Cloud Functions) goes further: you upload a small function that runs only when something triggers it, and you pay per call, by the millisecond. It is excellent for spiky, occasional jobs (send the trip receipt when a ride ends) and expensive for constant heavy work (the matching engine that runs every second).

Three ways to pay for the same server

On-demand: pay by the second, no commitment; the most flexible and the highest price. **Reserved or savings plans:** commit to one or three years of use for discounts of up to about 72% on AWS; right for the base load you are sure of. **Spot:** spare capacity at up to about 90% off, but the provider can take it back with two

minutes’ notice; right for work that can pause, such as retraining the ETA model overnight. A sensible bill mixes all three (Figure 4.41).

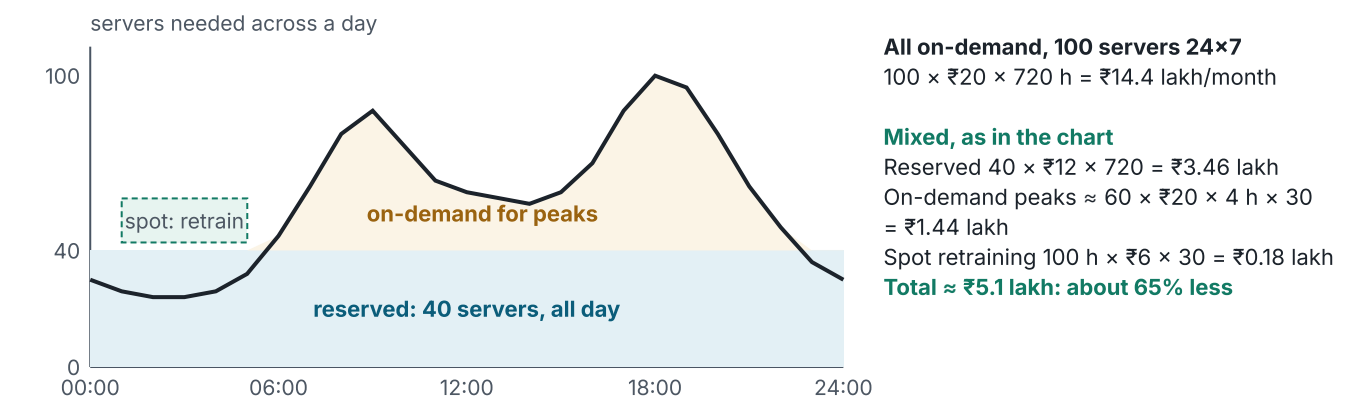


Figure 4.41 • One day of servers, bought three ways. Reserve the base that runs all day, rent the evening peak on demand, and push interruptible jobs to spot. The same capacity costs about a third. *Illustrative prices (₹20 an hour on demand, 40% off reserved, 70% off spot); published maximum discounts from AWS pricing pages, 2026.*

WHEN IT BREAKS

Bill shock: a forgotten test cluster or a runaway loop runs up lakhs in a weekend; budget alerts and tagging every server by team are basic hygiene. **A region goes down:** on 20 October 2025 a fault in AWS’s Virginia region took down many apps worldwide for hours; apps that ran in only one region went down with it. **Lock-in:** the deeper you use one provider’s special services, the costlier it is to leave, and moving data out (egress) is charged per GB.

IN THE INTERVIEW

“Our load swings 3× in a day and 10× on New Year’s Eve, so owning servers for the peak would waste most of them. I’d reserve the base, rent the peak, run model training on spot, and keep payments data in an Indian region. The decision I’d push for is a second region for the booking path, because a one-region outage stops every ride.”

PM decision	The cloud question behind it	Typical answer for a ride app
How reliable must it be?	One region or two? How fast must we recover?	Two regions for booking and payments; one for analytics
How fast can we ship?	Build on PaaS and managed databases, or run our own?	Managed services early; own only what gives an edge (matching)
What may it cost?	Cost per trip; reserved vs on-demand mix	Track cloud cost per 1,000 trips as a product metric
What do rules require?	Where may data live? (RBI rules for payments; the DPDP Act, 2023)	Payment data in Indian regions only

System design III: staying up when everyone books at once

Friday, 6 pm, rain in Bengaluru: ride requests triple in 20 minutes. New Year’s Eve at midnight: ten times a normal evening, in five minutes. The product team has to know three things before that night: which parts of the app must keep working, which may slow down, and what riders see when the system is overloaded. None of the ideas below needs a computer science degree. Every one of them already happens in a busy restaurant.

One restaurant, eight ideas

Picture a popular restaurant in Jayanagar on Diwali night. It faces the same problems as a ride app at midnight on 31 December, and it has found the same solutions. Keep this table in mind; each idea in this section is one row of it.

Idea	The restaurant on Diwali night	The ride app on New Year's Eve
Scaling out	Add more cooks at more stoves, not one faster cook	Add more identical servers
Load balancer	The head waiter seats each group at a free table, and skips the section whose waiter is on a break	Sends each request to a healthy server with room
Autoscaling	Calling in extra staff when the queue forms; they take 30 minutes to arrive, so for Diwali they are booked in advance	Adds servers when load rises; pre-scales for known peaks
Consistency vs availability	Never promise the last table to two families; the menu board can be a little out of date	Never give one driver two rides; the map of cars can be 10 s old
SLO and error budget	"At most 1 order in 100 may be late"; after too many late orders, no new dishes until the kitchen catches up	"99.9% of bookings work"; when the budget is spent, stop shipping features
Graceful degradation	When swamped, stop serving the elaborate thali and keep the main menu going	Switch off promos and extras so booking keeps working
Retries, circuit breaker	Guests shouting "where's my food?" every minute slow the kitchen further; when the tandoor breaks, stop taking tandoor orders	Retries pile load on a sick service; a breaker stops calling it for a while
Monolith vs microservices	One big kitchen doing everything, or a food court of separate stalls	One program, or many small services

The load, in numbers

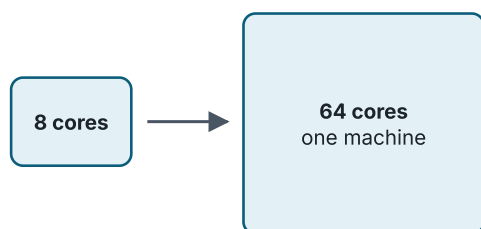
Say a ride app across India handles 5,000 requests a second on a normal Friday peak: fare checks, bookings, status updates. New Year's Eve brings 10×: **50,000 requests a second**. Separately, every online driver's phone sends its location every 4 seconds. 2 lakh drivers ÷ 4 s = **50,000 location updates a second**, all night, whatever riders do. These two streams need different care: bookings must be exactly right; locations just need to be handled in huge volume. (Numbers illustrative.)

1. Scaling up or scaling out

Everyday picture. A railway station before a festival can make one ticket clerk faster, or open more ticket windows. One fast clerk still has two hands, and if that clerk falls ill, nobody gets a ticket. Ten windows can serve ten queues, and one closed window doesn't stop the station.

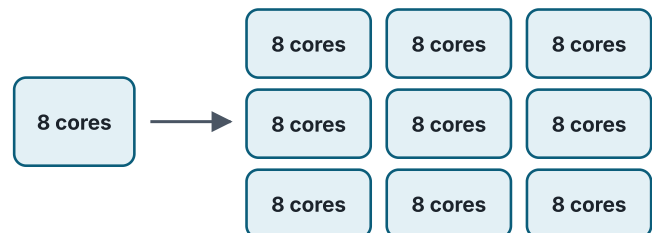
How it works. **Vertical scaling** (scaling up) buys a bigger machine: more processors, more memory. It needs no change to the software, but there is a biggest machine you can buy, and one machine is one point of failure. **Horizontal scaling** (scaling out) adds more machines of the same size and spreads the work (Figure 4.42). It has no ceiling, but it only works if any machine can handle any request. That needs the service to be **stateless**: like a bank, where any cashier can serve you because your balance is in the bank's central ledger, not in one cashier's head. The rider's trip details therefore live in a shared database, not on whichever server took the booking.

Vertical: a bigger machine



- + simple: no code change
- a ceiling: the biggest box
- one failure takes all down
- downtime to upgrade

Horizontal: more machines



- + no ceiling; add or remove any time
- + one box fails, the rest carry on
- the service must be stateless
- needs a load balancer, more to watch

Figure 4.42 · Two ways to add capacity. Ride apps scale out: many identical servers behind a load balancer. The part that is hard to scale out, the database, is where most design effort goes. *Author's diagram.*

Real example. Stack Overflow, one of the busiest sites for programmers, famously ran for years on a small number of powerful servers: scaling up worked because its traffic was large but steady. A ride app's traffic

swings 10× in a night, so it scales out. **PM decision:** if a feature keeps information on one server (a half-filled form, a trip in progress), ask where it goes when that server disappears.

2. Load balancing

Everyday picture. A toll plaza on the Mumbai–Pune expressway. Cars don’t pick a lane at random: overhead lights show which booths are open, and a booth with a broken barrier shows a red cross, so nobody queues behind it.

How it works. A **load balancer** stands in front of the servers and sends each request to one with spare room (Figure 4.43). Simple rules work well: take turns (round robin), or pick the server with the fewest requests in hand. Every few seconds it sends each server a tiny “are you alive?” message, a **health check**. A server that doesn’t answer gets the red cross: no new traffic, and the others share its load. Riders never notice. Load balancers also spread traffic across **availability zones**: separate buildings, with separate power, in the same city.

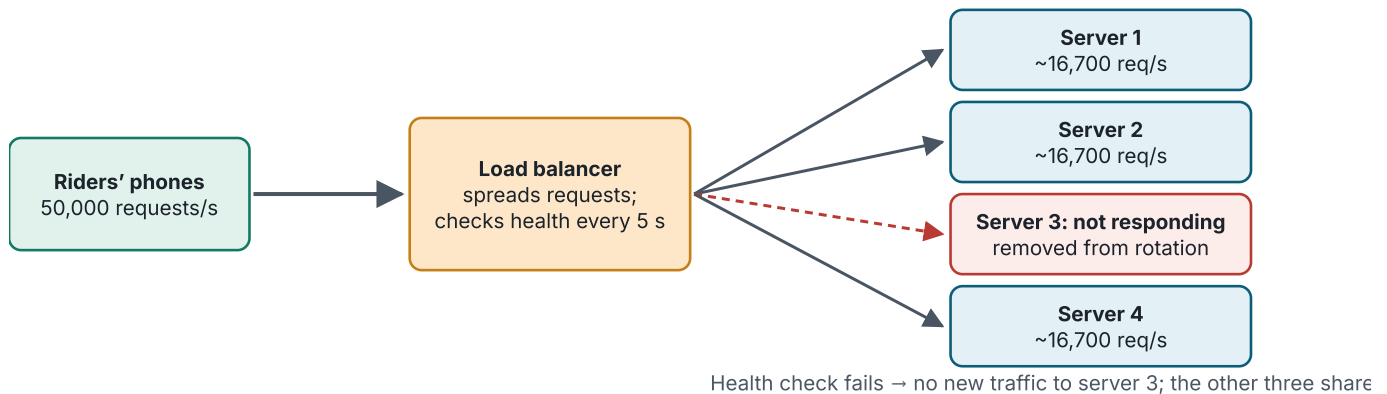


Figure 4.43 · A load balancer with a failed server. Health checks make one broken machine invisible to riders. *Illustrative numbers: 50,000 requests a second over four servers.*

Real example. In March 2021 a fire destroyed a building at OVHcloud’s data centre in Strasbourg, France. Websites that kept copies in other buildings stayed up; some that lived in one building went offline, and a few lost their data. **PM decision:** for the booking and payment path, insist on at least two zones; for an internal report, one may be enough.

3. Autoscaling, and why it arrives late

Everyday picture. When the restaurant’s queue suddenly reaches the pavement, the manager phones two extra waiters. They take 30 minutes to arrive, by which time half the queue has left. So for Diwali, the manager books extra staff a week ahead, and keeps one helper waiting in the back room for surprises.

How it works. **Autoscaling** adds servers when a signal, such as requests per second, crosses a line, and removes them when load falls, so you pay only for what you use. The sizing sum is simple: servers = peak requests per second ÷ requests one server handles, plus spare room. $50,000 \div 500 = 100$ servers; $\times 1.3$ for spare room = **130**. The catch is time: a new server takes 3–5 minutes to start, and the midnight spike arrives in five. Requests fail in the gap (Figure 4.44). For peaks you can see coming, scale on a schedule; for surprises, such as sudden rain, keep a small **warm pool** of servers already running.

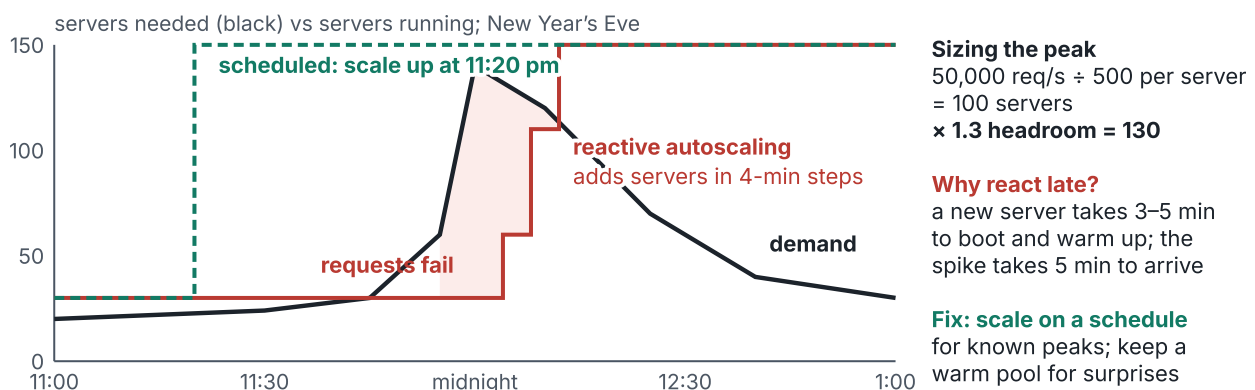


Figure 4.44 · Reactive autoscaling lags a sudden peak. The red area is requests that fail while new servers start. Scaling up at 11:20 pm on a schedule (green) closes the gap for a known peak. *Illustrative curve; start-up times typical of cloud servers, 2026.*

Real example. IRCTC's Tatkal window opens at 10 am for AC classes and 11 am for sleeper, every single day. Nobody waits for the rush to start adding capacity: the spike is on the calendar. The same is true of an IPL final, a Flipkart sale at midnight or a results day. **PM decision:** give engineering the calendar of your known peaks (launches, sales, festivals, marketing pushes) weeks ahead.

4. Right, or up? Consistency and availability, feature by feature

Everyday picture. Two counters of a cinema sell tickets for the same show, and the phone line between them goes dead. If both keep selling, seat F12 may be sold twice, and two families will argue in the dark. So seat sales must stop, or wait, until the line is back. But the poster showing the day's show timings can stay up even if it is an hour out of date. Nobody is harmed.

How it works. Large apps run in more than one data centre, and sometimes the link between them breaks for a few seconds: a **network partition**. The **CAP theorem** says that during a partition a system must choose: stay **consistent** (every answer correct and up to date, even if some requests must wait or be refused) or stay **available** (always answer, even if some answers are a few seconds old). The useful move is to choose per feature (Figure 4.45). A driver must never be assigned to two rides, so matching chooses consistency: better to wait two seconds than double-book. The cars on the map can be 10 seconds old, so they choose availability.

A network split: the Bengaluru and Chennai data centres can't talk for 30 seconds. Each feature must choose:

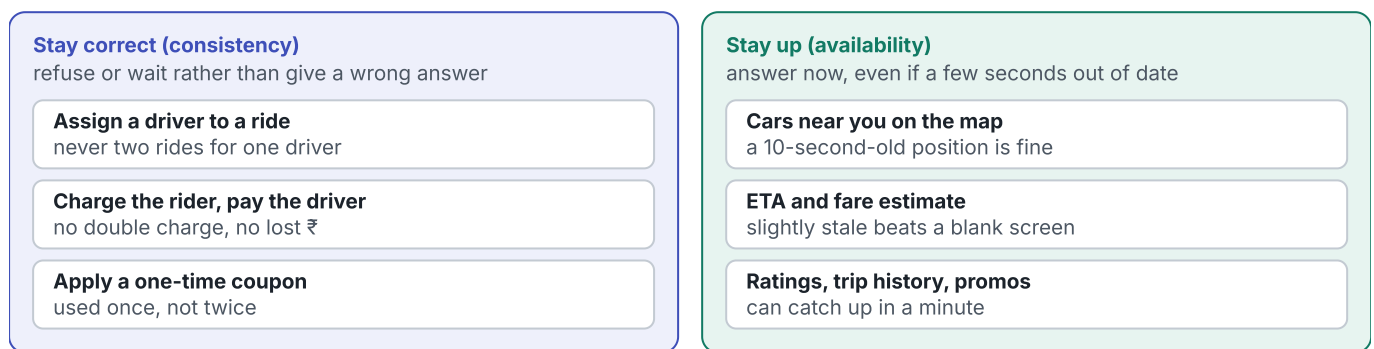


Figure 4.45 · Choosing per feature during a network split. Anything involving money or exclusive assignment must be right; anything that is only displayed can be slightly stale. *Author's classification for a typical ride app.*

Real examples. BookMyShow locks your chosen seats for a few minutes while you pay, so nobody else can buy them: consistency, bought with a short wait. An ATM whose link to the bank is down says “unable to process” instead of handing out cash it can't check: consistency again. WhatsApp's “last seen” is sometimes a little behind: availability, because a slightly old time hurts nobody. **PM decision:** for each feature, write one line: “if we're unsure, do we wait, or show the last known answer?”

5. Measuring reliability: SLIs, SLOs and error budgets

Everyday picture. A monthly mobile data pack. You get 45 GB for the month. Use it freely, stream what you like, but if 40 GB is gone by the 18th, you stop streaming video until the next cycle. Nobody argues about it, because the rule was agreed in advance.

How it works. An **SLI** (service level indicator) is what you measure: “share of booking requests answered successfully within 2 seconds”. An **SLO** (service level objective) is the target: “99.9% over 30 days”. The gap between the target and 100% is the **error budget**, the data pack of failure. $30 \text{ days} \times 24 \text{ hours} \times 60 \text{ minutes} = 43,200 \text{ minutes}$; 0.1% of that is **43.2 minutes** of failed bookings a month. While budget remains, the team ships features. When it runs out, it freezes feature releases and fixes reliability (Figure 4.46). One bad release that breaks bookings for 20 minutes spends $20 \div 43.2 = 46\%$ of the month's budget.

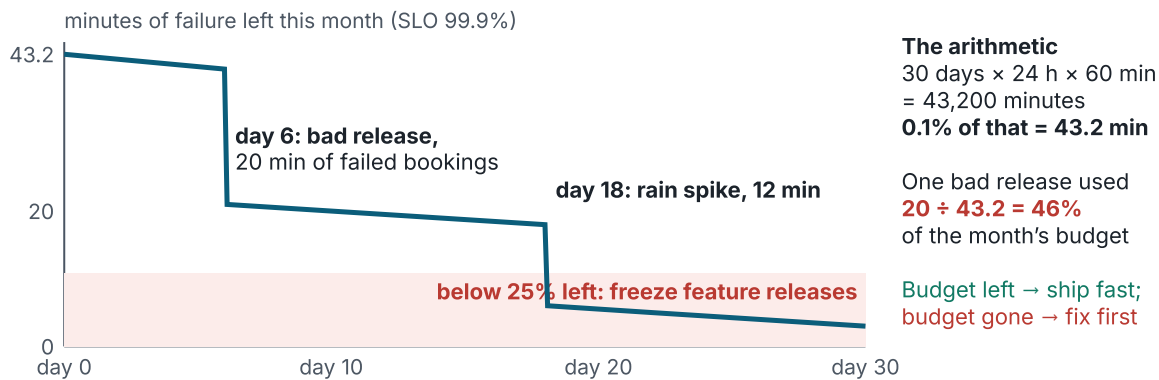


Figure 4.46 · An error budget over one month. Two incidents spent 86% of the budget by day 18; the team froze feature releases until reliability work was done. *Illustrative; the error-budget practice follows Google's Site Reliability Engineering (2016).*

Real example. NPCI publishes, month by month, how often each bank's UPI systems fail on their own side ("technical declines"). That share is an SLI anyone can read, and banks with high numbers are named in the tables. Google made the error-budget rule famous: product teams and reliability engineers agree the budget first, so the argument "speed or stability?" is settled by a number, not by who shouts louder. **PM decision:** choose the SLO from what riders feel (a booking confirmed within 2 seconds), not from what servers report. Each extra nine costs roughly ten times more effort, so don't ask for 99.99% where 99.5% would go unnoticed.

6. Failing gently: graceful degradation

Everyday picture. A power cut at home with an inverter. The inverter can't run everything, so it runs the fans, lights and the Wi-Fi router, not the air conditioner or the geyser. The house stays liveable because someone decided in advance which sockets matter.

How it works. **Graceful degradation** means that under overload the app gives up less important things so the core job keeps working. It is designed in advance as a ladder of switches (Figure 4.47), usually **kill switches**: settings an engineer can flip in seconds without a new app release. At 120% of planned load, switch off promo banners and recommendations. At 150%, show ETAs and fares from a saved copy up to 60 seconds old. At 200%, simplify matching to "nearest free driver". At 300%, queue new requests with an honest message ("high demand, about 3 minutes"). Only as a last resort, pause bookings in the worst zones.

Each step keeps the core job (get a rider a car) working by giving up something less important. Each is a switch built in advance.

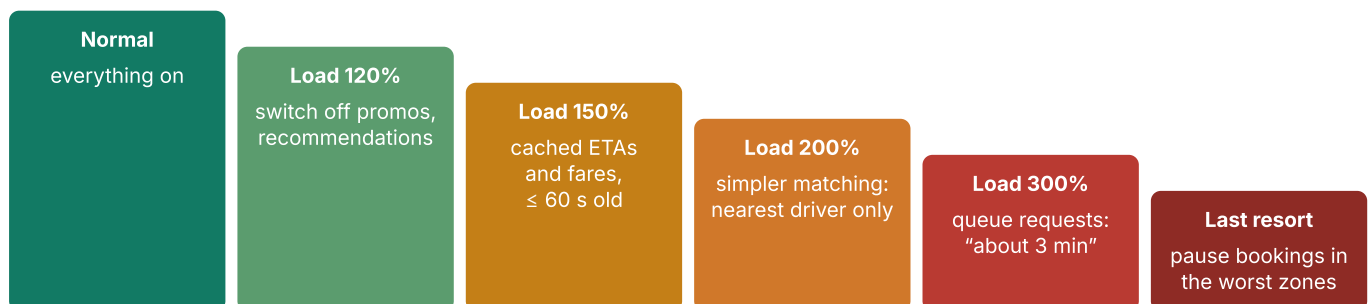


Figure 4.47 · A degradation ladder for a ride app. Each rung protects the core job by giving up something riders will miss less. *Author's example; the thresholds are illustrative.*

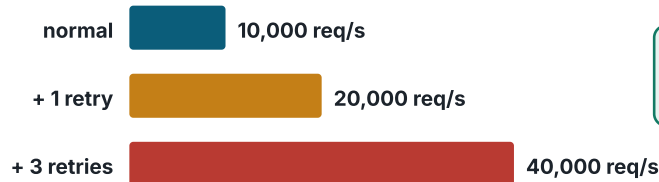
Real examples. When Netflix's personalisation system is struggling, the home screen falls back to a general list of popular titles instead of going blank; Netflix engineers have written about these fallbacks for years. IRCTC, at Tatkal time, puts some users in a waiting queue page rather than letting the whole site freeze. **PM decision:** the PM owns the ladder. Which feature is "the fan" and which is "the geyser" is a product call, and the switches must be rehearsed before the big night, because a switch nobody has tested often doesn't work.

7. Timeouts, retries and circuit breakers

Everyday picture. At midnight on New Year's Eve everyone calls their family at once. Calls fail, so everyone redials immediately, again and again, and the network jams even harder. The sensible thing is to wait a minute, and then a bit longer, before trying again. And in your home, when a faulty heater keeps overloading the circuit, the MCB (the switch in the fuse box) trips. It stops power to that circuit so the wiring doesn't burn, and after a while you switch it back on to see if the fault has cleared.

How it works. One ride request calls several internal services: pricing, maps, matching, payments. If pricing slows down and every caller **retries** at once, the struggling service gets several times its normal load at its weakest moment: a **retry storm**. With three retries each, 10,000 requests a second become 40,000 (Figure 4.48). Three defences. **Timeouts:** give up after a set time, shorter than the rider's patience. **Backoff:** wait longer before each retry (1 s, 2 s, 4 s), with a little randomness so callers don't all retry together. **Circuit breaker:** the MCB. When too many calls fail, the breaker "opens" and callers stop calling for a while, using a fallback instead (the last known fare, labelled "estimate"). After 30 seconds it lets a few calls through to test whether the service has recovered.

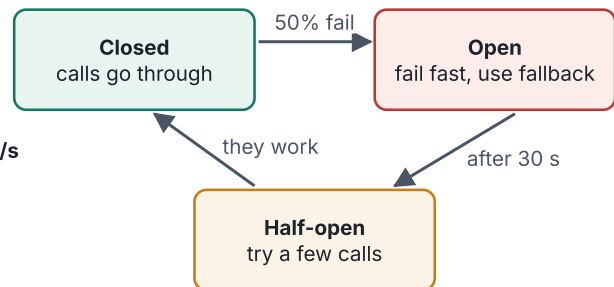
Retry storm: the pricing service slows down



Every caller retrying 3 times turns 10,000 requests into 40,000 at the moment the service is weakest.

Fix: back off (wait 1 s, 2 s, 4 s, plus randomness)

Circuit breaker: stop calling a failing service



Fallback while open: last known fare, marked "estimate".

Figure 4.48 · How retries multiply a failure, and how a circuit breaker stops it. The breaker trades a slightly worse answer (an estimated fare) for keeping the whole app up. *Illustrative numbers; the pattern follows Nygard, Release It! (2007).*

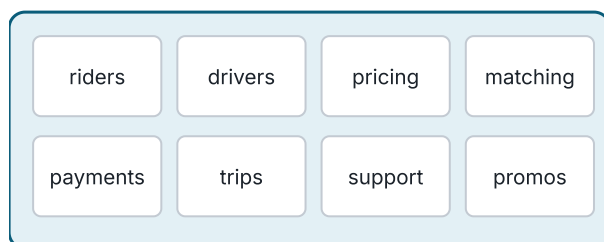
Real example. Netflix popularised circuit breakers in 2012 with an open-source tool called Hystrix, so that one slow service (say, ratings) couldn't freeze the whole app. Payment apps show the same idea to users: "Bank server is busy, please try after some time" is a circuit breaker speaking. **PM decision:** decide what the fallback looks like on screen. "Estimated fare", "ratings unavailable" and "try again in a minute" are product copy, not engineering details.

8. Monolith or microservices

Everyday picture. A family restaurant with one big kitchen: one head cook, one menu, one set of rules. Easy to run with ten staff. Now picture a mall food court: a dosa stall, a biryani stall, a juice stall, each with its own staff and hours. If the juice stall's blender breaks, you can still buy a dosa. But a "combo meal" across three stalls needs someone to coordinate, and when an order goes wrong, finding which stall caused it takes longer.

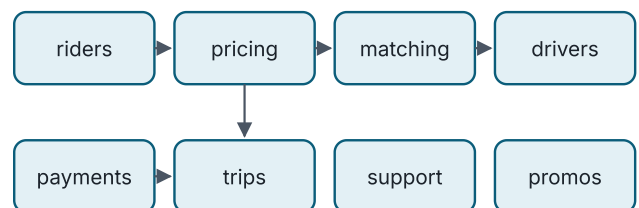
How it works. A **monolith** is one program that does everything, released as a single unit. It is simple and right for a young product. As teams grow it becomes a queue: 200 engineers wait for one release, and a bug in promotions can crash payments. **Microservices** split the product into small services, each owned and released by its own team, talking over the network (Figure 4.49). You can scale only the busy parts (matching at 9 am), and a crash stays local. The price is complexity: calls between services can fail, a bug can hide across ten services, and you need strong monitoring.

Monolith: one program, one deploy



- + simple to build, test and run at the start
- one bug in promos can crash payments
- 200 engineers queue for one release

Microservices: small services, each its own deploy



- + teams ship on their own; one crash stays local
- + scale only the busy parts (matching at peak)
- network calls fail; tracing a bug is harder
- needs strong tooling and on-call discipline

Figure 4.49 · One kitchen, or a food court. Microservices buy team speed and fault isolation at the cost of network complexity. Most young products should start with a well-organised monolith. *Author's diagram.*

Real examples. Uber moved from a monolith to microservices in the mid-2010s; by 2020 its engineers described about 2,200 critical services, and then grouped them into larger “domains” to tame the tangle. Going the other way, in 2023 the Amazon Prime Video team that monitors video quality merged one set of small services back into a single program and reported cutting that system’s running cost by about 90%. Neither answer is always right. **PM decision:** split when release queues and outages are slowing teams down, along the lines of how teams own the product, not because microservices sound modern.

Mechanism	Everyday picture	Where it breaks	The PM decision it drives
Scaling out	More ticket windows	Data kept on one server is lost with it	Where does each feature's information live?
Load balancer	Toll-plaza lane lights	Everything in one building	Two zones for booking and payments
Autoscaling	Extra waiters take 30 minutes	Late for sudden peaks; runaway bills	A calendar of known peaks; a cost ceiling
Consistency vs availability	Cinema seats vs the timings poster	Double-booked drivers, or a blank map	Wait, or show the last known answer?
SLO, error budget	The monthly data pack	A target nobody enforces	Agree the freeze rule before the incident
Graceful degradation	The inverter	Switches never rehearsed	Rank features; rehearse before big nights
Retries, breakers	Redialling; the MCB	A slowdown turned into an outage	What the fallback says on screen
Microservices	Kitchen vs food court	Too many, too early	Split along team lines when queues hurt

Putting it together: the New Year’s Eve plan

Like a restaurant preparing for Diwali, the whole plan is written weeks before the night. Each line below uses one of the eight ideas.

When	What the team does	Which idea, and the restaurant version
3 weeks before	Load test at 10× a normal Friday; fix whatever breaks first	Scaling out · a trial run with a full dining hall
1 week before	Rehearse every kill switch on the degradation ladder	Graceful degradation · test the inverter before the power cut
2 days before	Freeze feature releases; only fixes ship	Error budget · no new dishes on Diwali
11:20 pm	Scale to 130 servers on a schedule; warm pool ready	Autoscaling · extra waiters already on the floor
11:50 pm	Switch off promo banners and recommendations	Degradation · stop the elaborate thali early
Midnight	Watch booking success against the SLO; breakers ready to open on pricing and maps	SLO, circuit breaker · stop tandoor orders if it breaks
1:30 am	Scale down; switch features back on one at a time	Autoscaling · send the extra staff home
Next week	Blameless review: what failed, how much budget it used, what to change	SLO · the morning-after kitchen meeting

IN THE INTERVIEW · HOW SYSTEM DESIGN COMES UP FOR A PM

- **“Our app crashed on New Year’s Eve. What would you do?”** Prevention (scale up on a schedule, test at 10× load, rehearse the degradation ladder), response (flip kill switches, show an honest message), then learning (a blameless review against the SLO).
- **“Should we show the fare even if pricing is down?”** Yes, as a clearly labelled estimate from the saved copy; the booking then charges the confirmed fare. Available for display, consistent for money, like the cinema poster and the seat lock.
- **“How reliable should the app be?”** Pick a target riders would feel (bookings confirmed in 2 s, 99.9% of the time), turn it into a monthly budget (43 minutes), and agree what happens when it runs out.

Connect the dots

Four ideas ran through the cases, the products and the technology. A PM at a ride, maps or travel company meets all four in the same week, often on the same rainy evening.

Thread	In the cases	In the products	In the technology
Balance both sides	Cancellations were drivers voting on bad trips; the fix paid for the empty return	Uber's liquidity loop; Ola's loop running backwards; Rapido winning drivers first	Driver location pings are half the night's load; matching must stay consistent
Size for the peak	School vans break even at about 7 children a run; bike fleets sized for 9 am	Surge exists for the peak; airports and stations are separate markets	130 servers for midnight, pre-scaled at 11:20 pm; reserved base, on-demand peak
Test before you build	3 vans for 1 school; a fake-door "Plan my day" button; a stop rule for Dubai	Rapido's ladder climbs one tested rung at a time	Error budgets decide when to ship and when to fix
Know where the money is	₹110 Cr of UAE revenue made Dubai a beachhead, not a prize	Flights bring travellers, hotels pay; Maps earns on local intent; subscriptions earn per driver-day	Cloud cost per 1,000 trips as a product metric

BACK TO OUTER RING ROAD · A SKETCH ANSWER FOR THE RAINY-FRIDAY QUESTION

Pricing, matching or drivers? Start by splitting the failures. Are riders seeing "no drivers found" (supply is short), or are drivers accepting and cancelling (the trips are wrong for them)? Tonight both happen, but for different reasons.

Supply is short because the rain slows everything. A 35-minute trip now takes 55, so each driver completes fewer trips an hour; the same drivers serve about a third fewer riders. Surge at 1.8× is doing its job if it pulls drivers in from dry areas within 10–15 minutes. If it doesn't, the cap (2× base) may be too low to close the gap, and a longer honest wait is better than a spinning wheel.

Cancellations are the driver's maths. Jayanagar at 7 pm in the rain means a long crawl back. Split cancellations by drop area and time; if edge and long trips dominate, add a return premium and offer those trips first to drivers heading that way.

And the system must stay up. Rain triples requests in 20 minutes. Keep matching consistent, serve cached ETAs if pricing struggles, and switch off promos before bookings suffer.

Goal (North Star): completed trips per request tonight. **Safety checks (guardrails):** driver earnings per online hour; rider complaints about surge; booking success rate against the SLO. **Trade-off:** a higher price clears the queue but angers riders; a capped price keeps goodwill but leaves some riders without a car.

KEY IDEA · EVERY MOBILITY PROBLEM IS A PEAK PROBLEM

Drivers, bikes, school vans and servers are all capacity, and capacity is set by the busiest hour, not the average one. When you size a fleet, price a ride or plan a launch night, ask first: what happens at 9 am, at 6 pm in the rain, and at midnight on 31 December?